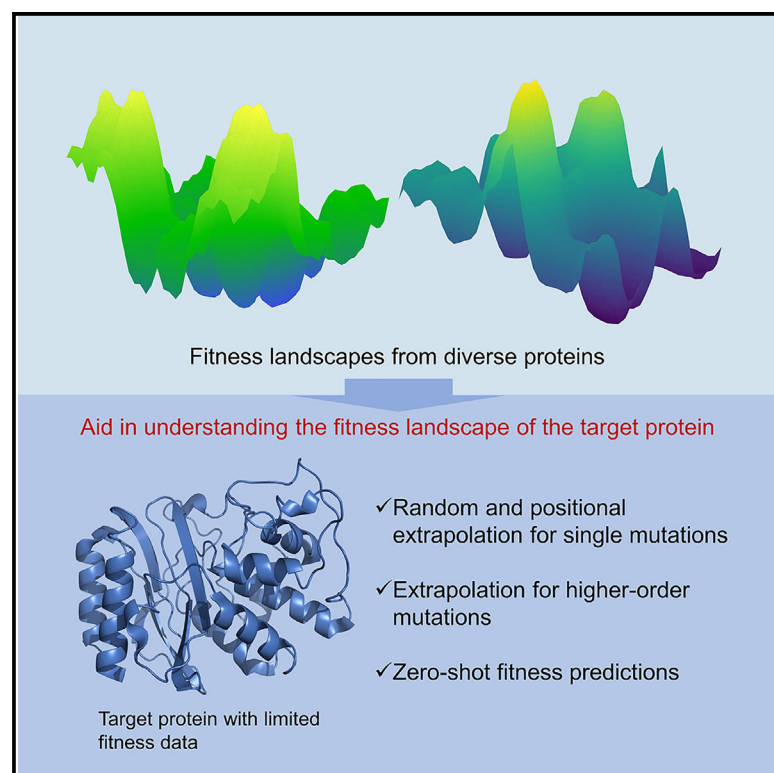


Learning protein fitness landscapes with deep mutational scanning data from multiple sources

Graphical abstract



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In brief

The fitness landscapes of different proteins may share similar underlying properties. This study demonstrates how deep mutational scanning data from various proteins can enhance the modeling of fitness landscape of a target protein. A deep learning model called GVP-MSA was developed, which considers both mutational structural environment and evolutionary context. Strong baselines and improved metrics were proposed to evaluate the model's performance. Overall, GVP-MSA may facilitate the exploration of fitness landscapes and epistatic interactions.

Highlights

- GVP-MSA effectively learns protein fitness landscape
- Both mutational structural environment and evolutionary context can be considered
- Fitness landscape learning can be transferred among different proteins
- Strong baselines and improved metrics are proposed for performance evaluation



Article

Learning protein fitness landscapes with deep mutational scanning data from multiple sources

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SUMMARY

One of the key points of machine learning-assisted directed evolution (MLDE) is the accurate learning of the fitness landscape, a conceptual mapping from sequence variants to the desired function. Here, we describe a multi-protein training scheme that leverages the existing deep mutational scanning data from diverse proteins to aid in understanding the fitness landscape of a new protein. Proof-of-concept trials are designed to validate this training scheme in three aspects: random and positional extrapolation for single-variant effects, zero-shot fitness predictions for new proteins, and extrapolation for higher-order variant effects from single-variant effects. Moreover, our study identified previously overlooked strong baselines, and their unexpectedly good performance brings our attention to the pitfalls of MLDE. Overall, these results may improve our understanding of the association between different protein fitness profiles and shed light on developing better machine learning-assisted approaches to the directed evolution of proteins. A record of this paper's transparent peer review process is included in the supplemental information.

INTRODUCTION

Protein engineering aims to create proteins with new or enhanced functions, which has the potential to solve a multitude of problems across the enzyme industry, medicine, and agriculture. One of the most effective approaches is directed evolution¹, an iterative search procedure based on artificial selection. After screening several random variants, the best variant is fixed as the parent sequence for the next round of directed evolution. Multiple rounds of mutations are performed until the desired performance is obtained. However, the greedy search strategy can trap the fitness into local optima and limit the sequence space considered. Hence, machine learning-assisted directed evolution (MLDE) was proposed and has been successful in facilitating the process of directed evolution to enhance enzyme function, fluorescent properties, and so on.^{2–4} The model learns the fitness landscape of the target protein through the training of experimental data and then selects a small and potential set of variants from the full combinatorial library for experimental validation. MLDE can explore a larger scope of sequence space *in*

silico, thus avoiding local fitness traps and alleviating the burden of experimental screening.

One of the key points of MLDE is accurate learning of the fitness landscape. Several deep mutational scanning (DMS) studies were performed to screen a mass of sequence variants and attempted to gain deeper insights into the mechanisms of variant effects and the topologies of fitness landscapes. Those experimental studies showed that most fitness landscapes are highly complex.^{5–11} In proteins, residues interact with each other in a dependent manner so that the effect of a higher-order variant is not always additive to their constituent single variants, known as the epistasis effect, which contributes to the ruggedness of the fitness landscape.^{5,12–14} In addition to the ruggedness, the sequence space is exponentially large; thus, exploring the full landscape is impossible. To get a glimpse of the fitness landscape, two main types of DMS experimental data have been generated in previous studies. The first type screens variant effects from a large sequence space, but they always contain only single or double variants or incomplete fitness information of higher-order variants. The second type involves the screening

of the full combination of a predefined small set of mutational positions, such as the fitness quantification of the full combination of the four epistatic residues of GB1 protein.⁸ Each category of the DMS datasets explores a small fraction of fitness landscapes from different perspectives.

Several methods have been developed to learn about the fitness landscape of proteins. They can be classified into unsupervised and supervised methods according to whether experimentally assayed fitness data are required for training. Unsupervised methods are applied to predict variant effects without labeled data. The existing zero-shot predictors can be further divided into two approaches. The first approach relies on learning the evolutionary contexts of the target protein sequences, such as EVmutation¹⁵ and DeepSequence.¹⁶ Since existing protein sequences are the results of long-term evolution, homologous sequences contain constraints to determine which mutations are tolerant and which are not. For EVmutation¹⁵ and DeepSequence,¹⁶ Markov random field (MRF) and deep generative model variational autoencoder (VAE) were used to capture the latent distribution in protein families, and the probability of sequence variants being sampled from that distribution was calculated. The log ratio of the variant and the wild-type probability are used to predict fitness of each sequence variant. Another type of approach of zero-shot predictor incorporates protein language models such as ESM-1b¹⁷ and ESM-1v.¹⁸ They are trained to learn to reconstruct the masked protein sequences based on their sequence contexts. The predicted likelihoods of the appearance of a particular mutational amino acid served as the output of the zero-shot predictors, which is consistent with the training goal of recovering word sequences. In contrast to learning from evolutionarily related sequences, they were trained on a diverse and large corpus of sequences, such as Uniref90,¹⁹ and learned the general patterns of sequence folding and functioning. Although ESM-MSA-1b²⁰ combined two paradigms, it predicted the masked amino acid in the context of evolutionarily related sequences and trained on large datasets of millions of multiple sequence alignments (MSAs), which fully leveraged both the evolutionary data of the target protein and patterns learned from other diverse protein family sequences.

In the second scenario, supervised methods are applied to predict variant effects through training on experimentally assayed data. For example, ECNet²¹ combined both pre-trained language models and evolutionary contexts from homologous sequences to learn the mapping from sequence variants to fitness. The protein language model in TAPE²² was utilized to extract the general patterns of sequence variants, and the MRF model CCMpred²³ was applied to learn features from MSA, in which pair-wise dependencies of residues were explicitly modeled. ECNet was shown to be more accurate than models without evolutionary context and can generalize to higher-order variants from the fitness data of low-order variants. Recently, Gelman et al.²⁴ examined four different neural network architectures and found that nonlinear module and parameter sharing across sequence positions were crucial for fitness landscape modeling. They constructed simple protein graphs with nodes representing features of residues and edges representing whether the distance between the two residues is within the specified threshold. It was shown that these graph convolutions incorporating protein structure information did not improve per-

formance over sequence-based models. They also created misspecified graphs that were unrelated to the protein structure and obtained similar performance as corrected graphs. These results may imply that we have faced pitfalls of machine learning analysis in protein fitness modeling, such as those pointed out by Patrick Riley.²⁵ To further explore these, more stringent performance evaluation schemes and more informative graph construction strategies are needed. More recently, Hsu et al.²⁶ developed a simple but effective ridge regression model that combined top-layer parameters amino acid feature and one feature from unsupervised modeling of the evolutionary data. The regression model showed good performance even when trained on fewer experimentally labeled fitness data.

Considering the complexity of the fitness landscape and the resource-intensive nature of the experiments, it is challenging to learn the mapping from sequence variants to fitness. In this work, we attempt to address this issue from a different perspective: to build a multi-protein model through different sources of fitness data to assist in the fitness landscape modeling of the target protein. First, we analyzed whether the mechanisms of variant effects and topologies of fitness landscapes of different proteins could be generalized. Several studies implied that there are underlying unified mechanisms of variant effects between disparate proteins. For example, Dunham et al.²⁷ combined DMS datasets from diverse studies to analyze the impact of protein positions and types of amino acids on variant effects. Gray et al.²⁸ used a stochastic gradient-boosting learning algorithm trained on several large-scale DMS datasets to predict the fitness of a new protein. Since the model describes the single amino acid substitution with 27 mutational, biophysical, and evolutionary features, it is limited to the prediction of single mutation effects, thus being unable to explore a larger mutational sequence space. Hoie et al.²⁹ predicted and interpreted general mechanisms of variant effects in terms of protein stability and sequence conservation through the analysis of DMS datasets. Here, we further investigated the commonality of disparate protein fitness landscapes. Besides the protein stability properties shared between disparate proteins, epistatic residues in different fitness landscapes tend to be structurally proximal. Second, we developed geometric vector perceptron (GVP)-MSA, a deep learning network to learn the fitness landscapes, in which a 3D equivariant graph neural network was used to extract features from protein structure and a pre-trained model MSA Transformer²⁰ was applied to embed MSA constraints, thus taking into account both the mutational structure environment and evolutionary contexts. We demonstrated that GVP-MSA can learn the underlying variant effects rather than merely intrinsic data bias of specific protein fitness landscapes through the modification of train/test split strategies. Third, we used the multi-protein learning framework to achieve the message integration of different protein DMS datasets and help reason about the fitness landscape of a new protein. We validated the generalization ability in three scenarios: random and positional extrapolation of single-variant effects, zero-shot fitness predictions for new proteins, and extrapolation for higher-order variant effects from single-variant effects. Finally, we further visualized and interpreted the internal representations of the model to explore what it has learned and how it may help us better understand the mechanisms of variant effects.

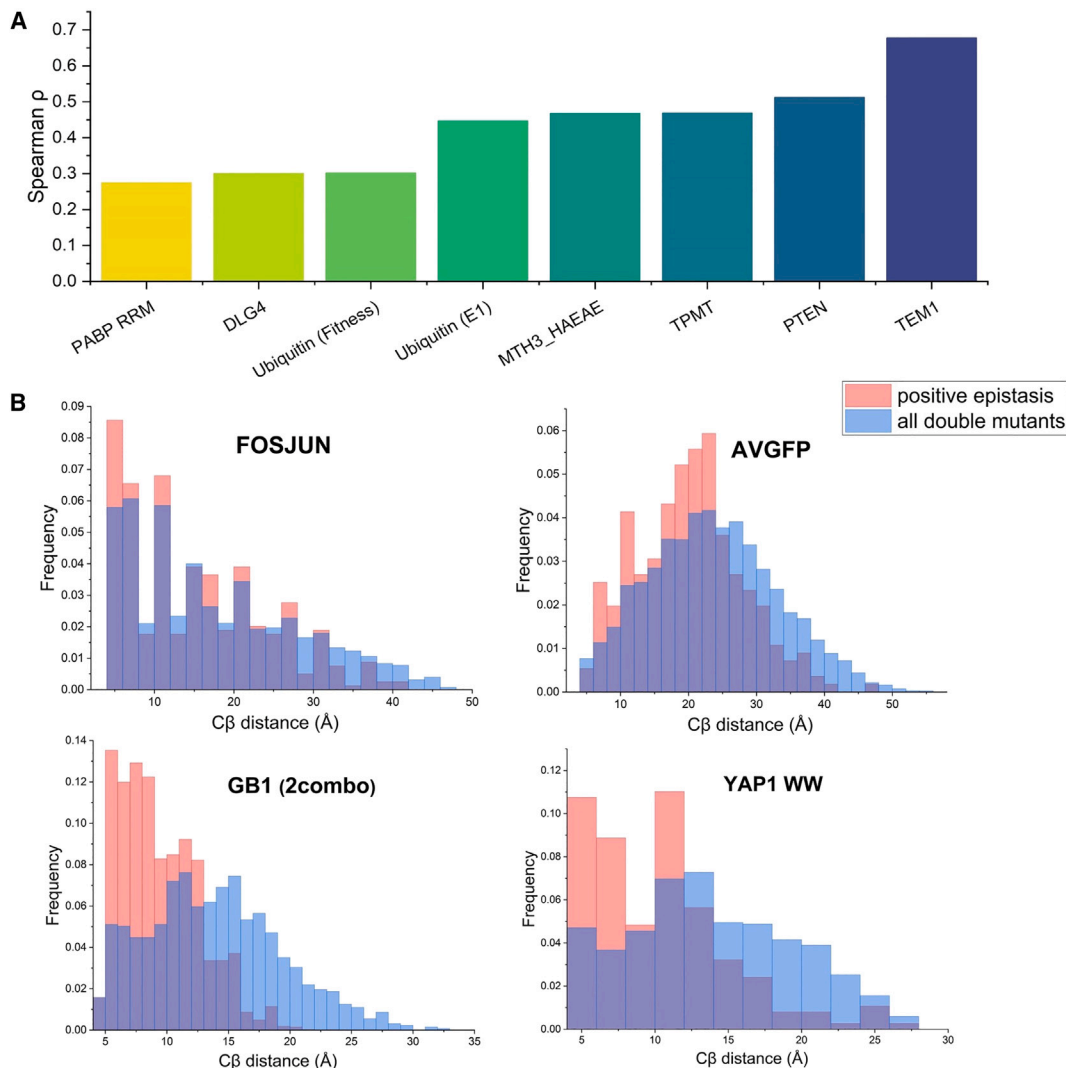


Figure 1. The motivation of our multi-protein scheme for fitness landscape learning

(A) Thermostability correlates with fitness in various deep mutational scanning experiments. The bar plot shows the Spearman correlation between fitness and thermostability calculated by Rosetta.

(B) The constituent single mutations of double variants with positive epistasis effects tend to be structurally proximal. Each histogram shows the distribution of distances between residues measured by the distance of their β -carbon atoms (C β). Pink histograms show the distances of double variants with positive epistasis effects, whereas blue ones show the distances of all double variants as a reference.

RESULTS

Fitness landscape of disparate proteins sharing similar properties

The fitness data reflect different properties according to different experimental systems, such as enzyme activity, binding affinity with peptides or molecules, fluorescence effects, and so on. However, the basis of a protein's function is its stability. Mutations that disrupt protein structure will result in a complete loss of function. Hence, this observation implies that there are general physicochemical properties of stable folding across diverse protein fitness landscapes. To verify this, we applied the Cartesian $\Delta\Delta G^{30,31}$ protocols of Rosetta³² to predict each variant's thermostability changes after mutation, which are quantified by the

relative Gibbs free energy change ($\Delta\Delta G$). Since large proteins require much longer to perform the biophysical calculations of Rosetta, several DMS datasets with relatively short protein sequence lengths were selected and studied. We found that the negative of predicted $\Delta\Delta G$ values correlated with protein fitness data, with Spearman's correlation coefficients ranging from 0.28 to 0.7 (Figure 1A). These results demonstrate that the fitness landscape measured for various protein functions contains implicit thermostability features.

In addition to thermostability, epistasis effects are pervasive in protein fitness landscapes, which complicates the prediction of higher-order mutations. It was found that the epistatic interaction can be partly explained by the interaction of structurally proximal residues.¹⁰ Also, the epistasis information was successfully used

to predict protein contacts and structure.^{33,34} To verify this, we collected four DMS datasets that contain the fitness of variants with both single and double amino acid substitutions. First, we calculated the epistasis values of each double variant and selected variants with epistasis effects (see the [STAR_Methods](#) section). Then, we measured distances between residues via their distances from the β -carbon atoms ($C\beta$) and analyzed the distribution of distances for double variants with or without epistatic interaction ([Figure 1B](#)). It can be found that the constituent single mutations of double variants with positive epistasis effects tend to be located structurally close to each other. This trend is shared across four disparate protein fitness landscapes.

Besides the commonality among fitness landscapes in terms of thermostability properties and epistasis rules, relationships between variant effects and their latent distribution of homologous sequences are also common. As evidence of this, unsupervised methods such as SIFT,³⁵ EVmutation,¹⁵ and DeepSequence¹⁶ have successfully predicted variant effects based on the analysis of evolutionarily related sequences. Different proteins share the same rules, that is, mutations tend to be deleterious if they never occur in their homologous sequences, and vice versa. These results motivated us to construct a multi-protein framework and leverage diverse DMS data to facilitate the fitness landscape modeling of target proteins.

GVP-MSA combines protein structure and evolutionary information to predict variant effects

It has been shown that the variant effect is highly dependent on its context in protein structure.³⁶ The incorporation of protein structure can help identify structurally proximal residues, even if they are remote in the primary sequence, which provides accurate information about the physicochemical environment around the mutational residues. In addition, evolutionary information from MSAs reflecting natural selection pressure can reveal clues as to which mutations are allowed.^{15,16} Therefore, we built a machine learning model, GVP-MSA, that combined both structural and evolutionary information to learn the mapping from sequence variants to protein fitness.

Our model comprises two main architectures to extract features from MSA and protein structures, respectively. Existing methods to embed MSA representation include using position-specific substitution matrices^{35,37} or Hidden Markov models,^{21,38} which consider residue sites independently or only the pair-wise residual interactions. However, inspired by the accuracy improvement of methods that consider higher-order dependencies within residues when predicting the protein structure or the variant effects,^{16,39} we also attempt to capture the higher-order dependencies. Recently, it has been shown that the protein language model trained on a large amount of sequence data can learn general patterns of protein sequences. The representation learned can be semantically rich and improve multiple downstream tasks such as secondary classification, structure prediction, and sequence-function mapping.^{17,40} Accordingly, here, we use a protein language model, MSA Transformer,²⁰ to capture the evolutionary context from MSA. MSA Transformer takes a set of aligned evolutionarily related sequences as input and learns the common patterns of protein families. Its architecture interleaves row and column attention across the sequence alignment, which facilitates the extraction

of higher-order dependencies within residues. All parameters of the pre-training model are fixed during the training of our model since we regard it as a feature extraction tool.

In addition to the evolutionary feature, protein structure information was also integrated into our model in a 3D equivariant way. Powered by the increasing number of protein structures in the Protein Data Bank (PDB) and accurate protein backbone structure predictors available (e.g., AlphaFold2³⁹ and RosettaFold⁴¹), protein structure information is much easier to access. Equivariant neural networks (ENNs) have emerged as a new class of methods for representing 3D structure and are shown to possess the ability to improve performance on multiple tasks, such as protein-protein interaction, protein folding quality assessment, computational protein design, and so on.^{42,43} Therefore, our model utilized GVP⁴⁴ layers to directly operate on rotation-equivariant vectors calculated from protein backbone structure, thereby avoiding the annoying transformation of local coordinate systems and using the information provided in the 3D coordinates more effectively. In addition to the rotation-invariant scalar features on each amino acid, geometric features represented by rotation-equivariant vectors were also included and updated during each layer. Thus, our model can represent and reason about the protein structure in an efficient 3D equivariant way. In GVP-MSA, we jointly combined the information from evolutionary contexts and protein structure to learn the mapping from sequence variants to fitness. The embedded MSA information was concatenated into the node representation when constructing the protein graph, thus allowing message communication between the MSA and protein structure. The MSA information is derived from the wild-type sequence and remains unchanged for different variants. To include mutation information, one-hot encoded representations of the wild-type and variant sequences are added to the node features of the protein graph. For protein-specific training, fitness data from the specific mutagenesis study was used to update the parameters of the neural network ([Figure 2A](#)).

In the previous section, we have shown that fitness data from disparate proteins may possess and share the same properties. Based on these commonalities, we built a multi-protein framework to simultaneously learn the fitness landscape from different protein mutagenesis data. All parameters on the bottom of the model were shared between proteins, and each protein had specific top-layer parameters. Within this framework, we want to investigate whether variant effects can be generalized and transferred between different proteins; hence, proteins with a large amount of known experimental data can be used to facilitate the engineering of a new protein of interest with a limited amount of experimental data. For this purpose, we have devised the following three test scenarios based on different numbers of training data available: random and positional extrapolation of single-variant effects, zero-shot fitness predictions for new proteins, and extrapolation for higher-order variant effects from single-variant effects ([Figure 2B](#)).

Accurate modeling of the protein-specific fitness landscape and performing multi-protein training

We investigated the performance of GVP-MSA on multiple benchmarking DMS datasets, where the fitness of single variants was measured based on different protein functions. We

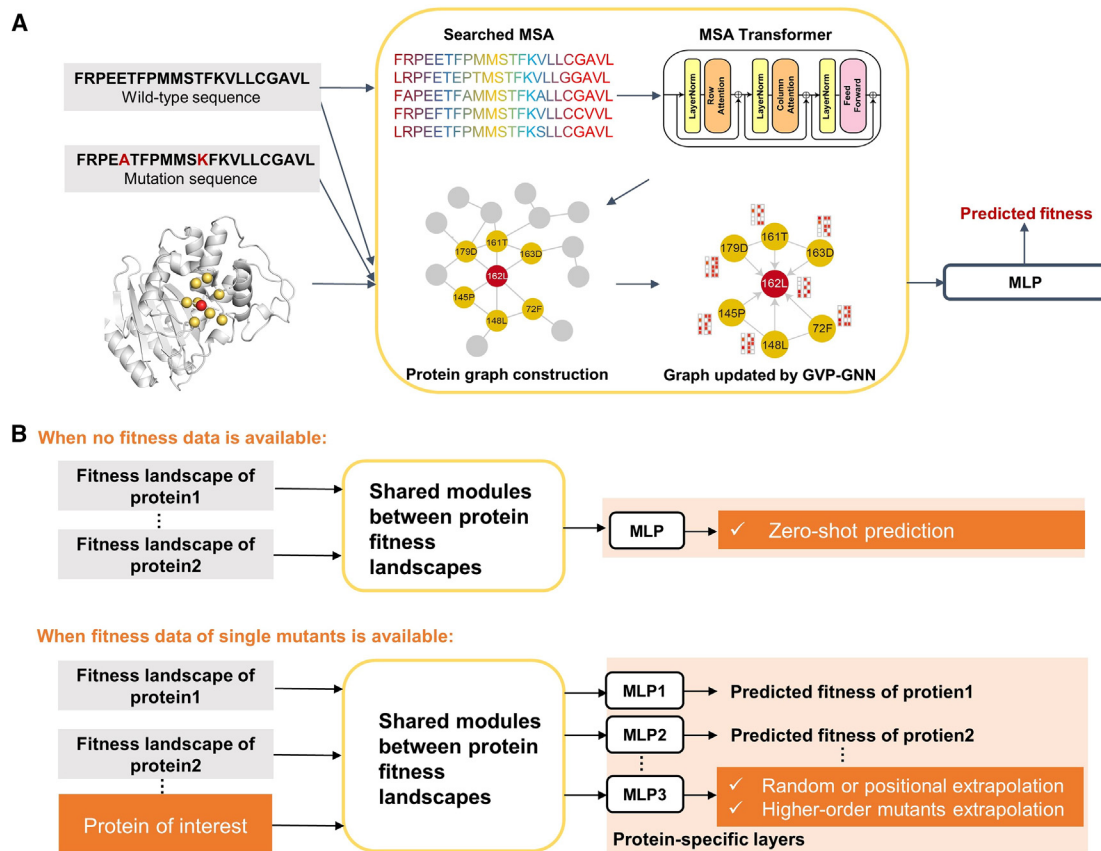


Figure 2. Overview of our supervised learning framework

(A) Overview of the protein-specific model, GVP-MSA. The model architecture takes as input the wild-type sequence, mutation sequence, and structure of the wild-type protein. First, MSA is searched based on the wild-type sequence from the protein sequence database Uniref90. The protein language model, MSA Transformer, is used to extract evolutionary contexts from MSA, and the output representation is added to generate the node features of the protein graph. Then, two layers of GVP-GNN are used to update the graphs. The evolutionary and structural contexts are combined to learn the fitness landscape of a specific protein. (B) Conceptual illustration of our multi-protein framework in different training scenarios. When no fitness data for the target protein were available, data from different protein fitness landscapes were combined to train one single model. When some fitness data points are available, the multi-task framework is constructed, and each fitness landscape will have its specific MLP layer on top of the model. The performance of this training strategy was tested in terms of random and positional extrapolation of protein-specific modeling and higher-order variants extrapolation. MLP: multiple layer perceptron.

managed to assess its capability for learning the mapping of variants to fitness in both protein-specific and multi-protein training scenarios.

First, we built a simple average model to demonstrate the pitfalls of conventional training schemes in terms of fitness landscape data. The fitness landscape has intrinsic data tendencies, and variant effects can be similar between different amino acid substitutions at the same mutation position. Therefore, we developed a simple baseline model called “Average”, where the predicted fitness is equal to the average of variant effects at the same mutation position in the training dataset. This was inspired by the good performance of a protein fitness landscape model based merely on one-hot encoded residue type feature.²⁶ Additionally, it has been explored in several experimental studies that mutations on the same position tend to show uniform effects.^{45,46} We found that the simple averaging model performs unexpectedly well when splitting the train/test dataset randomly (Figure 3C), indicating that this evaluation scheme is not suitable and can be overoptimistic to assess the model’s ability to learn

variant effects. To address this problem, we also evaluated the model’s performance in a position-specific scenario, where DMS datasets were split based on the sequence position to ensure that mutation positions observed in the training datasets did not overlap with those observed in the testing datasets (Figures 3A and 3B). In the position-specific scenario, we aimed to assess if the model could learn the underlying mechanisms of variant effects instead of the intrinsic data bias of a specific protein fitness landscape.

Therefore, we examined the model’s capability in two training schemes: protein-specific and multi-protein. In the protein-specific training scheme, a specific model was trained and tested for each protein fitness DMS dataset. In the multi-protein training scheme, we performed multi-task training to simultaneously leverage fitness data from different DMS studies. We aimed to investigate whether the performance of protein-specific modeling can be improved by incorporating DMS data from other proteins. The training was conducted as follows: First, we trained a reference multi-task model with several DMS

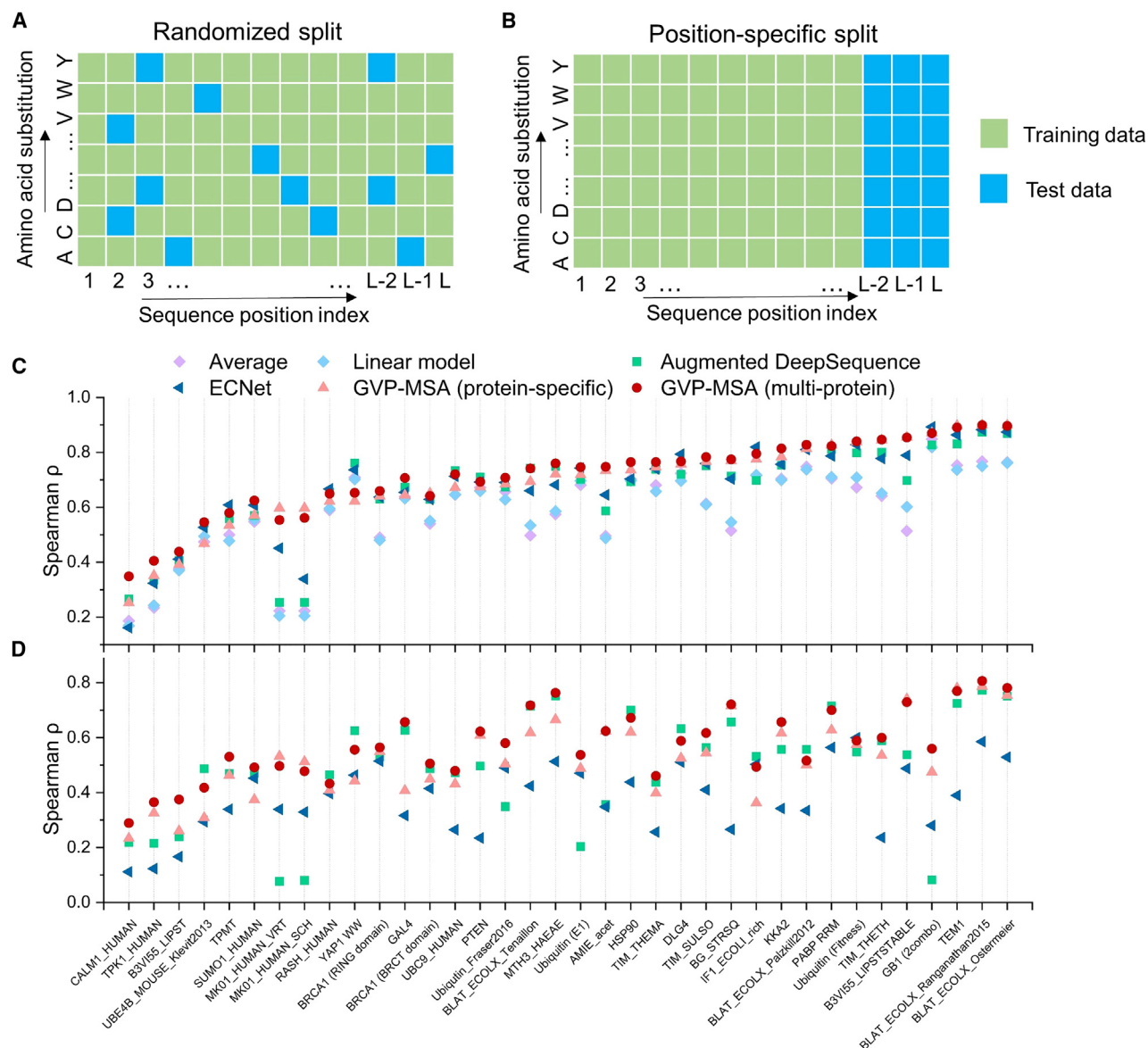


Figure 3. Comparison of protein-specific fitness landscape learning models based on different data split schemes

(A and B) Schematic visualizations of two data split schemes. The x axis represents the index of position in the protein sequence, and the y axis represents different choices of amino acids that are mutated. Two data split schemes were considered: randomized split (A) and position-specific split (B).

(C and D) Model performance of protein-specific learning based on the randomized split (C) and position-specific split (D) schemes. See also, Figures S1.

datasets, in which the parameters of the bottom model are shared and the parameters of the top model are specific for different proteins. Then, the model was finetuned by their own fitness data based on the reference multi-protein model (training details in STAR Methods). We also analyzed the number of DMS datasets that should be incorporated into the multi-protein training procedure. We gradually added more DMS datasets as training data by selecting datasets with higher test metrics in protein-specific training. This was intended to give preference to datasets with less noisy or unexplainable data. As we increased the number of training data, we observed an improvement in the model's performance. However, this improvement stopped at nine training datasets. This number can be consid-

ered a trade-off between the quantity and quality of training data (Figure S1A).

To validate GVP-MSA model and the multi-protein training strategy, we first compared it with ECNet,²¹ which combines evolutionary contexts extracted from MSA by direct coupling analysis²³ and global contexts embedded in the protein sequence pre-training model in TAPE²². ECNet has been shown to perform better than a wide range of methods, including unsupervised methods such as EVMutation,¹⁵ DeepSequence,¹⁶ and Autoregressive,⁴⁷ as well as supervised methods such as Envision,²⁸ sequences represented by Doc2Vec⁴⁸ or TAPE, and UniRep.⁴⁹ We have collected 36 DMS datasets where variants with single amino acid substitutions were screened based on

different protein functions. In the random train/test splitting scenario, GVP-MSA achieved a higher Spearman's correlation coefficient than ECNet for most of the proteins (21 of the 36 proteins are statistically significant; one-sided Mann-Whitney U tests $p < 0.01$; [Figures 3C and 3D](#)). Moreover, we observed that the simple averaging model also shows good prediction performance. Hence, we split the dataset in a position-specific way to test whether models have predictive power at a particular position when all fitness information for that position is removed from the training dataset. In this position-specific splitting scenario, we found that our model consistently outperformed ECNet on Spearman's correlation coefficients for 30 of the 36 protein fitness datasets. These results indicate our model's capability of modeling fitness landscapes rather than merely learning the data bias of the intrinsic fitness distribution of a specific protein. Furthermore, we found that the multi-protein training strategy can improve the performance on both random and position-specific splitting schemes (Spearman's correlation coefficients were improved in 25 of the 36 proteins when splitting datasets randomly and in 29 of the 36 proteins when splitting based on mutation site; one-sided Mann-Whitney U tests $p < 0.01$; [Figures 3C and 3D](#)). These results highlight the effectiveness of the multi-protein training paradigm.

Next, we compared our model with the ridge regression model developed by Hsu et al.,²⁶ which combined evolutionary information with experimentally assayed data and outperformed more sophisticated methods such as UniRep,⁴⁹ eUniRep regression,⁵⁰ fine-tuned ESM-1b,¹⁷ the integrated Potts model,⁵¹ and TLMutation.⁵² We compared our model with their augmented DeepSequence, which achieved the highest performance among other methods. Our model GVP-MSA, trained with the multi-protein strategy, showed improved performance for most proteins on both randomized and position-specific splitting schemes (Spearman's correlation coefficients were improved in 30 of the 36 proteins when splitting datasets randomly and in 25 of the 36 proteins when splitting based on mutation site; one-sided Mann-Whitney U tests $p < 0.01$; [Figures 3C and 3D](#)). We also evaluated their l2-regularized linear regression with one-hot encoded representation. Similar results were obtained with the Average baseline in the randomized split scheme, and it showed no predictive ability in the position-specific split scheme as it yielded the same predicted value for different variants, which is consistent with their proof that l2-regularization helps to learn the mutation tendency of each position.

From our perspective, the above tasks tested are of limited practical value since 80% of the single mutation datasets were needed to train a model. Instead, we regard it as an approach to validate the model's capability of learning fitness landscapes and the effectiveness of the multi-protein training strategy. We also designed the following generalization tasks that correspond to real-world application scenarios: predicting variant effects without training data and extrapolating to higher-order mutation effects from single variants.

Multi-protein training of GVP-MSA is an effective zero-shot predictor

Most mutations in protein sequences are deleterious for their function. During the process of MLDE, these protein variants with low fitness will provide less information when used to train

to prioritize high-performing variants.⁵³ Therefore, it is highly desirable to establish an effective zero-shot predictor that can select informative variants for experimental screening in the first round of directed evolution. Accordingly, models that can fully leverage the informative training data will better navigate the fitness landscape and facilitate protein engineering.

We compared our model with a broad range of zero-shot predictors, including protein language models such as ESM-1v¹⁸ and ESM-1b,¹⁷ and the models based on the learning of homologous sequence distributions such as EVmutation¹⁵ and DeepSequence.¹⁶ We also compared our model with ESM-MSA-1b, a protein language model trained on MSA that combines the advantages of both the protein language model and evolutionary information. Our model used a new scheme to make zero-shot predictions, where a unified variant effect predictor was trained on a large amount of fitness data from diverse proteins and then generalized to the target protein without experimental data. To simulate real-world-directed evolution, in which multiple amino acid substitutions are always taken into account, we collected six DMS datasets containing both single and higher-order variants. The test data sets are diverse, with fitness being measured based on different functions ranging from enzyme activity to protein-protein binding affinity. In addition, six DMS datasets with only single variants were included in the training set. We employed a leave-one-protein-out training approach to evaluate the model's capability on new proteins (details were included in [STAR Methods](#)). We found that our model showed stable performance among different protein datasets and achieved the best performance when calculating the mean value among all tested datasets. Specifically, GVP-MSA outperformed other methods in proteins with small MSA effective sizes, such as avGFP and GB1. Our methods were less affected by the low quality of MSA, and although no MSA was provided, the GVP-Mut model still had the ability to perform zero-shot prediction. The results were in line with our expectations because methods such as EVMutation and DeepSequence learn latent protein family distribution from MSA, thus heavily relying on the quality of MSA. However, our method utilizes the fitness data of other proteins to train the model, thereby reducing the model's requirements for the quality of MSA. In addition to Spearman's correlation coefficient, we applied another ranking measure known as normalized discounted cumulative gain (NDCG). In accordance with MLDE's goal of prioritizing high-fitness variants of proteins, the NDCG metric gives more weight to iterative mutation, of high-fitness variants than low-fitness variants ([STAR Methods](#)). As shown in [Figure S2](#), quantitatively similar results were observed when performance was measured by NDCG. Furthermore, the accuracy of variant effects prediction will be increased if fitness data are available for proteins whose sequences are similar to those of the test protein. For example, to predict the fitness of avGFP, we collected datasets of fitness landscapes for three orthologous fluorescent proteins: amacGFP (*Aequorea macrodactyla*, Hydrozoa), cgreGFP (*Clytia gregaria*, Hydrozoa), and ppluGFP2 (*Pontellina plumata*, Copepoda), with 18%, 59%, and 82% sequence divergence from avGFP, respectively.⁵⁴ When adding them to the training dataset, the Spearman's correlation coefficient improved significantly ([Figure 4B](#); two-sided Mann-Whitney U tests, $p < 0.01$). Moreover, we observed that the ensembled results from different

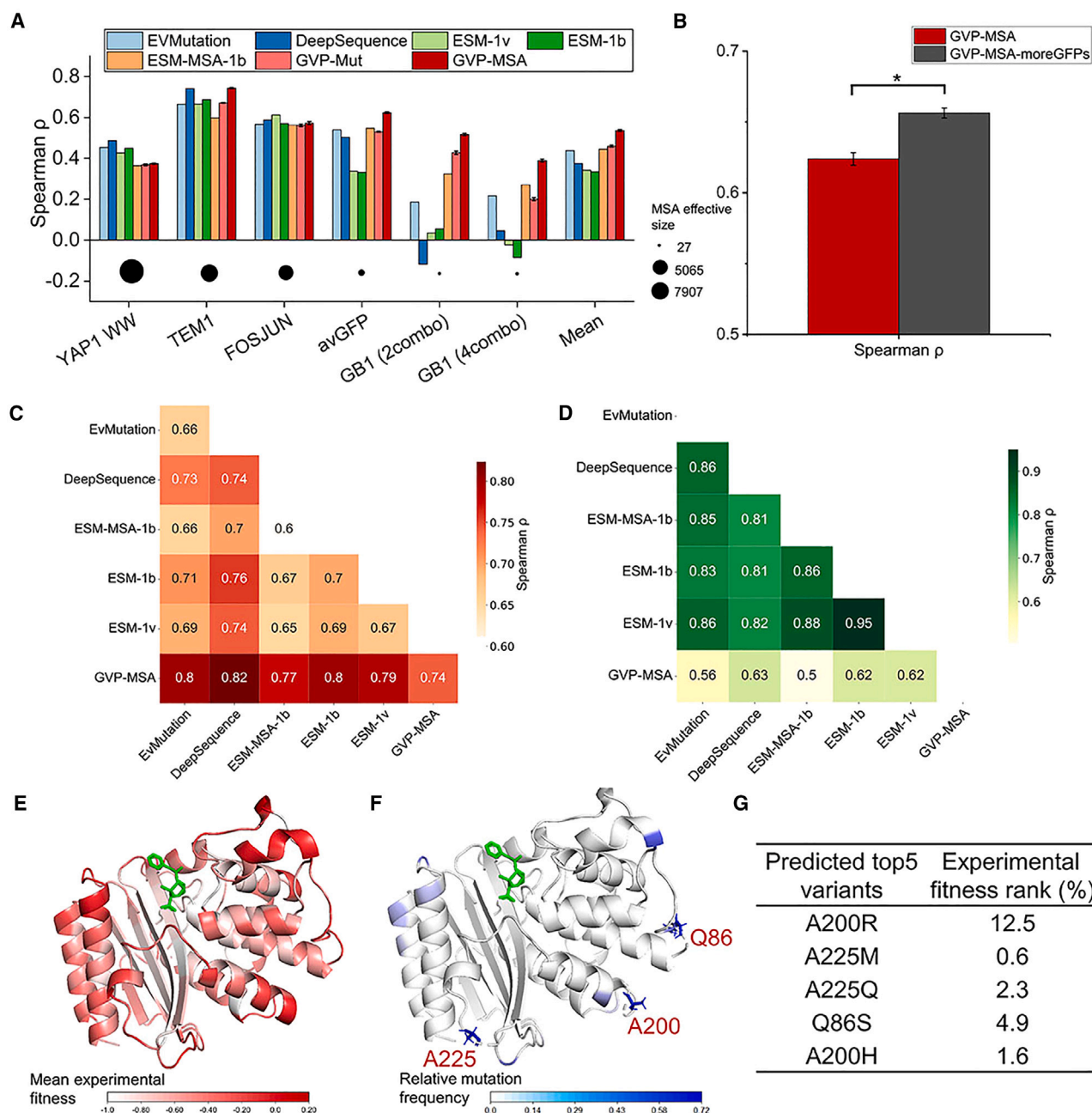


Figure 4. GVP-MSA is an effective zero-shot predictor

(A) Comparisons with five unsupervised models on six DMS dataset. The “Mean” column is the mean Spearman’s correlation coefficient value among six datasets. The size of the circle indicates the number of MSA effective sizes. Error bars on the bar plots represent the standard deviation of 15 replicate tests.

(B) Comparisons to training with more similar DMS datasets. When trained with fitness data of other fluorescent proteins, GVP-MSA can improve the predictive performance of avGFP dataset (two-sided Mann-Whitney U tests $p = 3.4 \times 10^{-6}$; sample size $n = 15$). Asterisk (*) indicates that $p < 0.01$ in two-sided Mann-Whitney U tests that the two methods have significantly different performances. Error bars on the bar plots represent the standard deviation of 15 replicate tests.

(C) The performance of pair-wise ensemble methods. The diagonal value shows the Spearman’s correlation coefficient of the predicted fitness with experimental fitness from a single method.

(D) The pair-wise Spearman’s correlation coefficient of the predicted fitness of the TEM1 dataset from different methods.

(E) Mean experimental fitness values on each position mapped onto the TEM1 structure. The catalytic active site is highlighted in green.

(F) Structural visualization of TEM1. Mutations are colored by relative frequency in the top 3% of predicted fitness. The catalytic active site is highlighted in green. The top three residues by mutation count are shown as sticks.

(G) The table shows the predicted top five variants and their rank in the experimentally assayed fitness. See also, [Figures S2 and S3](#).

zero-shot predictors are generally better than the results from single methods. Taking dataset TEM1 as an example, our method showed better ensemble performance than other methods' pair-wise ensemble results (Figure 4C). To account for this observation, we calculated the pair-wise Spearman's correlation coefficient between the results predicted by different zero-shot predictors and found that our method showed less correlation with others. Additionally, we found that the protein language models ESM-1b and ESM-1v exhibit the strongest correlation between their predictions, which is in accordance with their similar methodological basis (Figure 4D). Predicted results of other test datasets exhibited similar tendencies (Figure S3). These results suggested that our unique training strategy of GVP-MSA captured the effective properties of variant effects that were different and complementary to those of other unsupervised methods. Then, we examined the predicted results to determine which types of variants tend to be prioritized by our method. We found that the variants predicted to have high fitness were enriched at the positions that are solvent-exposed and away from the active site, which was consistent with the experimentally assayed results (Figures 4E and 4F). Moreover, we checked the variants that predicted the top five and indeed found that these variants showed higher fitness values in experiments (Figure 4G). These results suggest the feasibility of the multi-protein training strategy of GVP-MSA for zero-shot prediction and identifying informative variants for experiment screening in the first round of directed evolution.

GVP-MSA can extrapolate to higher-order variants effects from single variants

The mutational sequence space grows exponentially as the number of mutations increases. For instance, describing the entire fitness landscape of single variants for a protein with 100 amino acids requires 1,900 (19×100) variants. However, searching all combinations of double variants requires nearly 2,000,000 ($c_{100}^2 \times (20 - 1)^2$) variants. Experimentally screening all mutational combinations is impossible. It has long been recognized that higher-order variants exhibit epistasis effects, meaning that variants containing two or more amino acid substitutions from wild-type sequences can have non-additive mutational effects.^{12,13} It is defined as positive epistasis effects when the experimentally assayed fitness of higher-order variants is higher than the addition of their constituent single variants, and vice versa (Figure 5A). Furthermore, positive-sign epistasis occurs when the positive epistasis effect is strong enough to transform the higher-order variant into an advantageous one from its constituent deleterious single variants (Figure 5B). From the perspective of Darwinian evolution, a deleterious variant has a very low probability of fixing, and thus, the positive-sign epistasis can constrain the accessibility of direct pathways of protein adaptation.^{8,55,56} This phenomenon in natural protein adaptation was pervasive and was well illustrated by Weinreich et al.⁵⁵ Among 120 mutational trajectories arriving at a particular beneficial 5-point variant of the beta-lactamase, 102 are inaccessible to Darwinian selection. Protein-directed evolution simulates the natural adaptation process to obtain variants with higher desired functions by iterative mutation, which means that this process will also be constrained. Higher-order variants with positive-sign epistasis effects will

have a low probability of being obtained since their parent variants will not be fixed. Therefore, it is highly required to develop an effective model to predict higher-order variant effects and detect positive-sign epistasis effects that may hinder protein adaptation.

We collected seven diverse DMS datasets containing both single variants and higher-order variants: YAP1 WW,¹¹ TEM1,⁷ FOSJUN,⁶ avGFP,⁹ GB1 (2combo),¹⁰ GB1 (4combo),⁸ and severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2).^{57,58} Among them, GB1 (4combo) contains nearly all mutational combinations of four structural proximal residues, and TEM1 contains only consecutive amino acid substitutions, whereas the other datasets contain double or more variants from random substitution mutation sequences. The variant effects data of SARS-CoV-2 were collected from two different studies. Specifically, the single-variant effects data of SARS-CoV-2 were generated by Starr et al.,⁵⁷ in which all single amino acid substitutions to the receptor-binding domain (RBD) of the SARS-CoV-2 spike protein were generated, and their binding affinities to human cell-surface protein angiotensin-converting enzyme 2 (ACE2) receptor were experimentally measured. The higher-order variant effects data of SARS-CoV-2 were collected from Taft et al.,⁵⁸ in which variants with one to 14 amino acid substitutions were evaluated to determine whether they would bind to the human ACE2. Table S1 provides detailed information on these data sets.

It has been demonstrated historically that most higher-order variants have additive effects of their constituent single variants on protein function,⁵⁹ and we found that most DMS datasets show similar full-zero epistasis distribution (Figure 5C; Figure S4). Therefore, before testing more sophisticated models, we devised a simple baseline model called "Addition," where the predicted fitness of higher-order variants is set to the addition of their respective single-variant effects (STAR_Methods). We also trained GVP-MSA for extrapolation from single variants to higher-order variants using two strategies referred to as "Protein-specific" and "Multi-protein." Protein-specific means directly training on single variants of the target protein and testing its performance on higher-order variants of that protein, whereas the Multi-protein strategy incorporates the single and higher-order variants of other protein fitness landscapes by training them in a multi-task way (STAR_Methods). We compared our models with ECNet, which was shown to be able to generalize to higher-order variants from single variants. In addition to ECNet, we also compared it with the l2-regularized linear model with one-hot encoded feature and the augmented DeepSequence model from Hsu et al.²⁶ Unexpectedly, the simple additive model outperformed all the sophisticated methods in five of the six data sets (Figure 5D), indicating the pervasiveness of the additive effect and the sparsity of the epistasis effect. Additionally, GVP-MSA trained with the multi-protein strategy outperformed GVP-MSA trained with the protein-specific strategy, ECNet, and models from Hsu et al. in four of six data sets, again emphasizing the effectiveness of the multi-task learning strategy. Moreover, we also tested each method's ability to predict the binding affinity of ACE2 RBD with SARS-CoV-2 higher-order variants based on the single-variant effects data. Considering that the label of higher-order variants is a binary value, we used the receiver operating

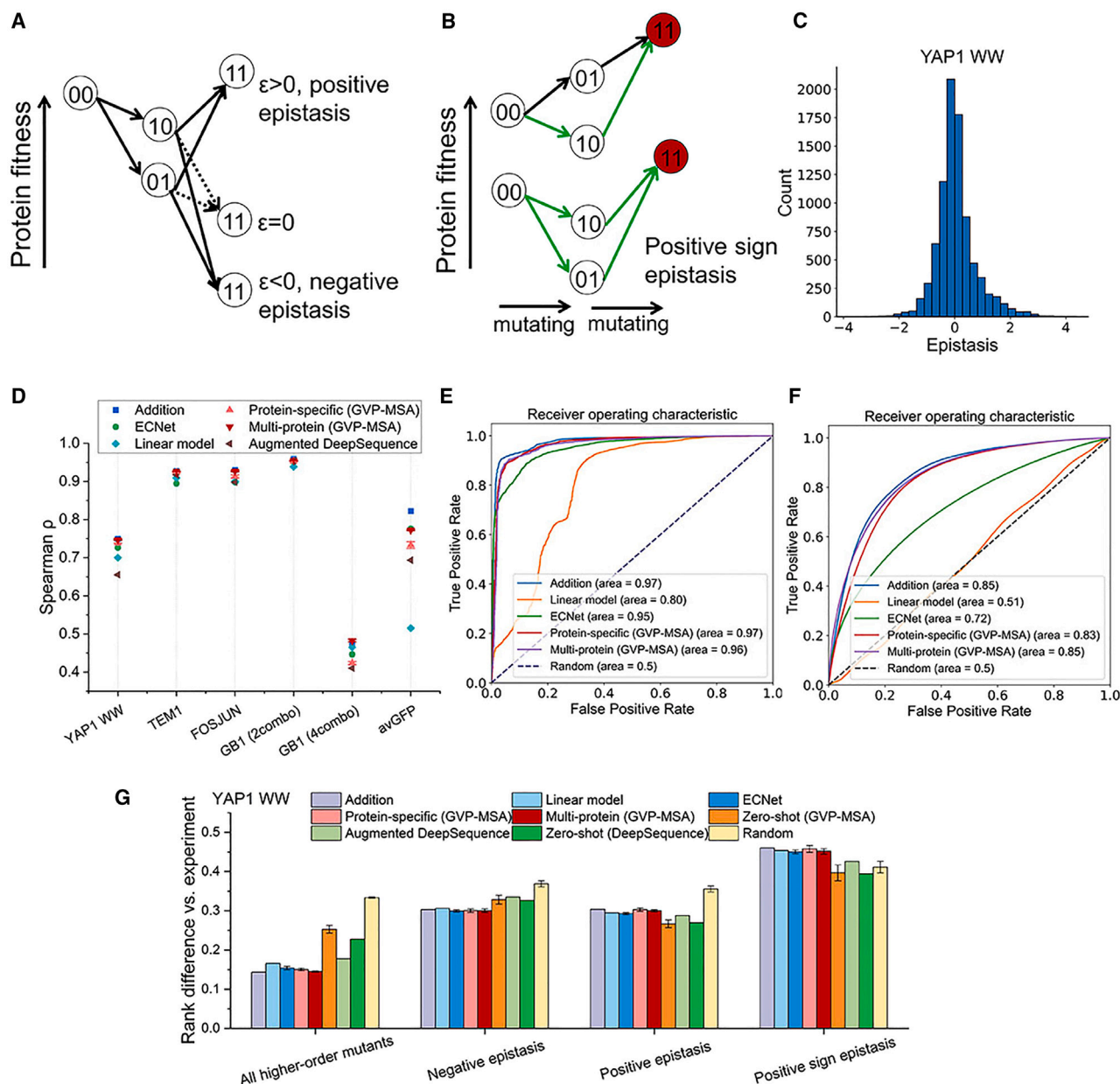


Figure 5. Extrapolation to higher-order mutation effects from single mutation effects

(A) Schematic visualization of positive epistasis effects and negative epistasis effects

(B) Schematic visualization of positive-sign epistasis effects. The illustration shows possible intermediate stages for mutating from "00" to a beneficial double variant "11" (colored in red). Green arrows represent inaccessible pathways in the adaptation process.

(C) Histogram shows the distribution of epistasis value of dataset YAP1 WW.

(D) Prediction of higher-order variant effects by different models based on single-variant effects. The Spearman's correlation coefficients achieved by the method Addition were significantly higher than other methods in five of six data sets (the largest p values of all other methods in two-sided Mann-Whitney U tests: YAP1 WW: 3.4×10^{-6} , TEM1: 2.8×10^{-5} , FOSJUN: 3.4×10^{-6} , GB1 (2combo): 8.1×10^{-5} , avGFP: 3.4×10^{-6} ; sample size $n = 15$). When excluding the method Addition, the Spearman's correlation coefficients achieved by the method "Multi-protein (GVP-MSA)" were significantly higher than other methods in four of six data sets (the largest p values of other methods in two-sided Mann-Whitney U tests: YAP1 WW: 2.5×10^{-3} , TEM1: 7.8×10^{-3} , FOSJUN: 4.9×10^{-4} , GB1 (4combo): 3.4×10^{-6} ; sample size $n = 15$).

(E and F) ROC curves illustrate the classification ability of models when predicting the binding affinity of ACE2 receptor-binding domain with SARS-CoV-2 higher-order variants based on the single-variant effects data. Prediction of mutation effects with fewer than six amino acid substitutions. Prediction of mutation effects with more than or equal to six amino acid substitutions.

(G) Comparisons of the mean absolute error (MAE) of rank difference between predictive results and experimental fitness in different groups of variants. The results in the "Random" method were obtained by averaging 500 random rankings. Lower rank differences indicate better ranking performance. Error bars on the bar plots represent standard deviation of the mean. See also, Figures S4 and S5.

characteristic curve to determine the model's performance in predicting higher-order variants. The simple additive method still demonstrated the best performance in both cases of higher-order variants containing fewer or more than six amino acid substitutions, which suggests that the additive effect may dominate the fitness landscape more than we have thought.

Next, we further investigated whether these models could detect epistasis effects. The higher-order variants were identified into three subgroups: negative epistasis effects, positive epistasis effects, and positive-sign epistasis effects. We compared each method's performance on different subgroups based on the mean absolute error (MAE) between the predicted rank and the experimentally assayed variants rank (see the [STAR Methods](#) section). It should be noted that Spearman's correlation coefficient was not used here because it is of more interest to evaluate the ability of the model to rank a certain small subgroup out of the entire variants population than to measure the ranking ability within the small group itself. We also compared zero-shot predictors that have no single-variant data for model training. As shown in [Figures 5G and S5](#), we found that when ranking variants with epistasis effects, the additive baseline model, which does not take epistasis effects into account, can sometimes perform better than sophisticated methods that utilize MSA information. Furthermore, the ranking performance can even be worse than the random ranking, and no method can consistently produce acceptable ranking results among different test data sets. These results raise the question of why we were unable to predict the epistasis effects of higher-order variants when MSA information was utilized. Since we have compared several models that incorporate MSA information through different methods and their MSA extraction methods have been demonstrated to be effective in predicting other tasks, such as protein contact maps, we believe that residual interaction information from MSA should be available to the model. Furthermore, previous works observed that for mutation positions that are structurally proximal, most double mutants do not display either form of epistasis, although double mutants with positive epistasis tend to be structurally proximal.¹⁰ Also, the negative epistasis effects are complex and may be explained by the threshold robustness model.⁶⁰ Proteins can tolerate some destabilizing mutations. However, as two destabilizing mutations occur concurrently, the stability threshold can be exhausted, and the double mutations become deleterious. Therefore, we hypothesize that the failure to detect epistasis effects may be due to the indirect relationship between the epistasis effects and the residual interaction information obtained from MSA. With the help of our multi-protein training scheme, we may be able to provide a possible solution to this problem by learning the general mechanisms of epistasis effects from diverse protein fitness landscapes. We also argue that the possible reasons for the difficulty of our model in detecting epistasis effects are the lack of sufficient training data and the complex epistatic interaction mechanisms. GVP-MSA, trained with multi-protein strategy, heavily relies on protein fitness data from different sources. However, the epistasis effects are highly sparse and can only be observed in a small proportion of higher-order variants. This poses a great challenge for data-driven methods.

Visualization and interpretation of the model

What did GVP-MSA learn? Did the model learn the underlying physicochemical principles or just learn data biases derived from the fitness datasets? How can it help us better understand the mechanisms of variant effects? To answer these questions, we visualized the model's latent representations in two-dimensional projections using t-distributed stochastic neighbor embedding (t-SNE). The latent representation of the test dataset TEM1 was extracted during the zero-shot prediction procedure so that none of the data points from that DMS study were used to train the model, allowing us to fully test how the model performed in out-of-distribution settings. The variants were projected according to their latent representations and colored by experimental fitness, calculated physicochemical properties, or the subtypes of residues before and after mutation. We found that variants with low fitness enrich on the top of the two-dimensional t-SNE representation ([Figure 6A](#)), which is associated with a higher value of $\Delta\Delta G$ changes calculated by the Cartesian application of Rosetta ([Figure 6B](#)). This observation is consistent with the knowledge that unstable variants tend to have a lower fitness value. Then, we investigated the relationship between the variant effects and the biophysical properties of mutation positions. Low relative accessible surface area of mutation positions is associated with low-fitness variants, which corresponds to our understanding that buried residues in protein structure tend to be intolerant of mutations ([Figure 6C](#)). The type of secondary structure also shows a pattern that mutations at the loop structure tend to contribute a higher fitness value ([Figure 6D](#)). In contrast to the relatively accessible surface area, the secondary structure shows lower relevance to fitness, and the potential reason is that the secondary structure considers only a subset of surrounding residues, whereas the relatively accessible surface area considers all surrounding residues. Overall, these results demonstrate that our model's internal representations indeed extract meaningful biophysical properties of mutations and sequence positions, as revealed by the marked patterns relating to experimentally assayed fitness values in the t-SNE space.

We then classified amino acids into typical subtypes and found that the subtypes before and after mutation also displayed distinct patterns ([Figures 6E and 6F](#)). Proline shows strong patterns with fitness, and it is clustered well for both residues before and after the mutation. A mutation in proline results in a decrease in protein fitness, and this observation agrees well with our knowledge about proline, which has a unique rigid cyclic ring compared with other residues. Non-polar amino acids are another notable group in the t-SNE representation of original residues. Their mutations tend to have a low-fitness value, cross-referencing with our relative accessible surface area's representation ([Figure 6C](#)), which reveals that these non-polar amino acids tend to be highly buried. Moreover, we found that mutating to aromatic residues or mutating from aromatic residues tends to have a low-fitness value. The aromatic amino acids contain benzyl groups with π -electron cloud and have a rigid and planar larger structure, which means mutating from aromatic amino acids to other residues may destroy the stability stemming from the π -electron cloud, whereas mutating to aromatic amino acids may introduce clashes with existing protein structure.

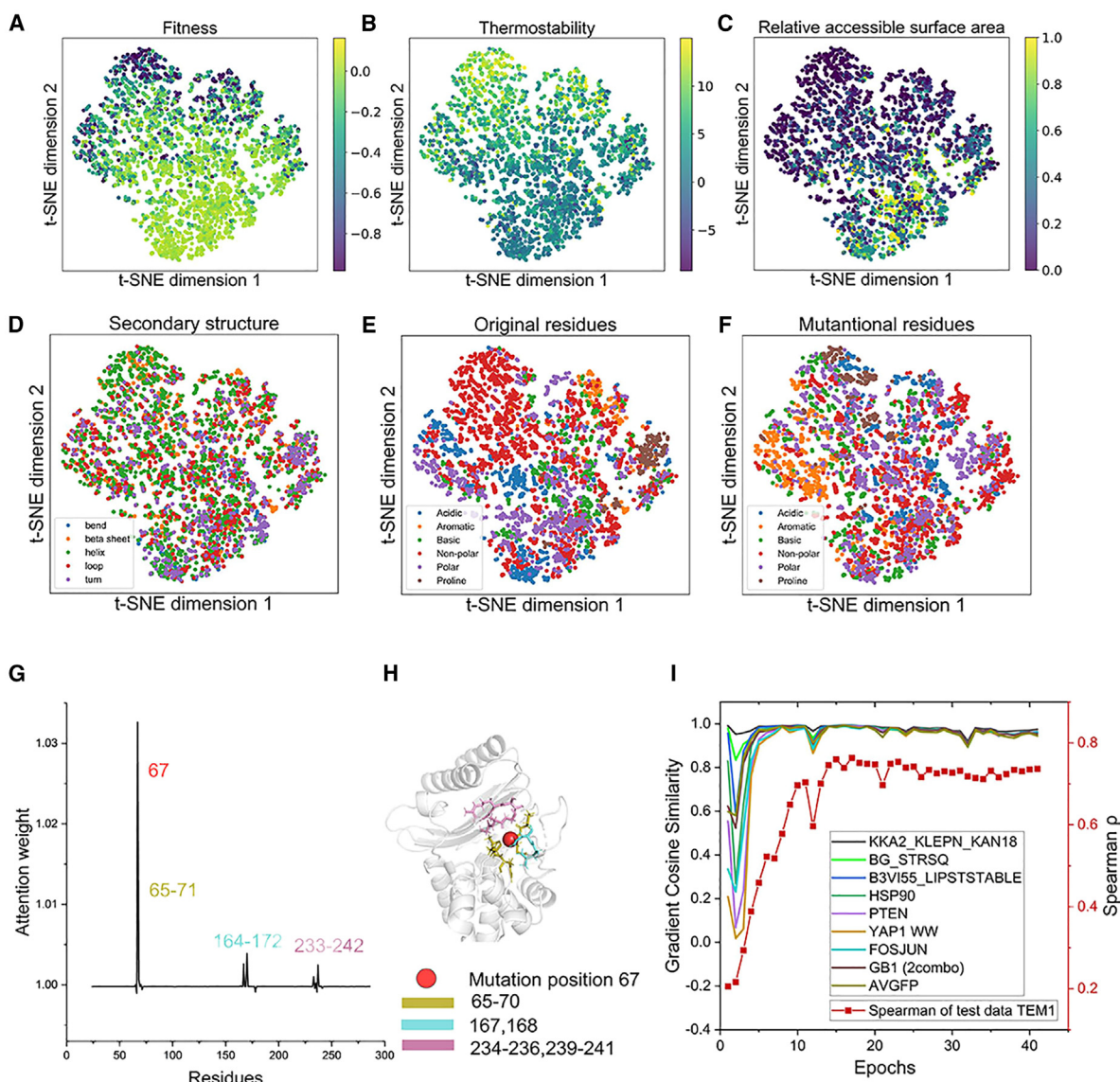


Figure 6. Neural network interpretation. Visualization of model's internal representations and gradients when performing zero-shot prediction of dataset TEM1.

(A–F) Two-dimensional t-SNE projection of the last internal layer of the neural network. Each point corresponds to one variant with single amino acid substitution, colored by the experimental fitness (A), the change of thermostability calculated by Rosetta (B), the relative accessible surface area of wild-type protein (C), the secondary structure of wild-type protein (D), and the amino acid types before (E) and (F) after mutation.

(G) Attention weight of mutation M67L when performing zero-shot prediction of TEM1 dataset.

(H) Structural identification of residues within 4 Å of mutation position 67.

(I) This chart shows how two different measures evolve as the training epochs. The left y axis represents the gradient similarities of different DMS training datasets with test dataset TEM1 of the last readout layer, whereas the right y axis represents the Spearman's correlation coefficient between the predicted and experimental fitness of the test dataset.

Overall, these findings support the hypothesis that GVP-MSA learns the underlying physicochemical properties of variant effects.

To validate that our model learns meaningful results, we also extracted the attention weight in the last readout layer to investigate whether the model paid more attention to important residues. Taking variant M67L as an example, we found that the model assigned more weight to the mutation residue and its

structural surrounding residues (Figures 6G and 6H). Then, we further investigated whether fitness data derived from disparate DMS datasets could be combined to serve as training resources. It has been reported that the gradient similarity measured in multi-task learning correlates well with task proximities and the model performance.^{61,62} During the optimization process, we calculated the cosine similarity of the gradient between disparate fitness datasets and found that the gradient of the fitness

datasets exhibited a similar direction as the training process progressed (Figure 6). Furthermore, we found that similar gradient trajectories, implying less interference between tasks, are associated with better test results.

DISCUSSION

In this work, we presented a deep learning model, GVP-MSA, that leveraged fitness data from different mutagenesis studies of diverse proteins and successfully facilitated the modeling of the fitness landscape of target proteins. First, we analyzed the commonality of the variant effect mechanisms and the similar patterns of epistasis effects behind different mutagenesis datasets, which demonstrated the feasibility of the proposed multi-protein training. We validated the model's capabilities in protein-specific scenarios by performing both random and mutational positional extrapolation. Then, we performed multi-task training and demonstrated that the model trained with fitness data from other proteins can facilitate protein-specific learning. Through benchmarking on 36 different DMS datasets, GVP-MSA outperformed other methods when tested in both random and positional extrapolation scenarios. Also, we performed zero-shot prediction and predicted variant effects of new proteins without experimentally assayed data by training a single model using diverse mutagenesis datasets of other proteins. Compared with other zero-shot predictors, our method showed consistently high performance, even for proteins with limited evolutionary data. Furthermore, we attempted to predict higher-order variant effects from single-variant effects and detect epistasis effects, especially for the positive-sign epistasis effects that may hinder protein adaptation. We found that the simple addition model outperformed all sophisticated models in predicting higher-order variant effects, and none of the compared methods yielded a consistently acceptable performance of detecting epistasis effects. Overall, GVP-MSA provides a framework for integrating data from different protein systems to facilitate fitness landscape exploration and deepen our understanding of the prediction of variant effects and epistatic interaction.

Moreover, our analysis identified some previously overlooked strong baseline models, including the position-specific averaging model for fitness prediction and the simple addition model for higher-order variant effect prediction, and these baseline models showed unexpectedly good performance in some scenarios. These results not only reveal the hidden biases in the data distribution but also bring our attention to the pitfalls of DMS data modeling. To detect and avoid the problems, we also proposed a series of practices, such as how to split the training and test data sets and how to set performance evaluation metrics more relevant to real-world scenarios, which will promote mining DMS data and exploring protein fitness space more efficiently.

It is important to note that there are potential issues when using DMS datasets from different sources as training data sets. In various DMS experiments, fitness was measured using different proxies such as the host growth rate, fluorescence levels in the cell, and steady-state abundance of protein variants. This can lead to different relationships between fitness and protein function. For instance, the *in vivo* fitness measured by host growth

rate has a sigmoidal function with *in vitro* protein function, as the fitness will saturate at some level of protein function. Additionally, different selective pressures can generate a different relationship between fitness and biochemical protein function.^{63,64} In our zero-shot prediction scenario, we attempted to make DMS datasets comparable based on the assumption that mutations resulting in a complete loss of function are comparable among different DMS datasets. However, this approach is still a compromise. In the multi-protein training scenario, we alleviate this issue through the multi-task training protocol. Under this protocol, different DMS data sets shared some parameters, whereas some specific parameters reflected their unique features.

We aim to learn the general features of fitness landscapes to assist in the process of directed evolution. For zero-shot prediction, the features we learned tend to be more related to protein stability. We trained an overall model to make predictions on new proteins, despite fitness being measured based on different functions in different protein systems. This can serve as an initial experiment design to filter out uninformative variants in the first round of directed evolution. Active learning has facilitated the development of better models in areas with fewer labeled training instances,^{65,66} showing prospects for addressing the limitations of resource-intensive experiments in directed evolution. Wittmann et al.⁵³ validated the strategy of running MLDE with a designed informative training set, referred to as "focused training MLDE" (ftMLDE). The ftMLDE strategy was shown to perform better in engineering new proteins compared with MLDE or traditional directed evolution. The pipeline of ftMLDE includes one or several rounds of informative training data design and few-shot learning. Future work can further explore how our training strategy, which employs a diversity of fitness data, can give new insights into these processes. A better understanding of the rules by which fitness landscapes are formed may facilitate these processes. Additionally, as the number of DMS datasets with higher-order variants increases, the mechanisms of the epistasis effects can be further explored, allowing us to access variants that are hindered by the natural adaptation process due to the epistasis effects. With the GVP-MSA model and multi-protein training scheme, we expect to facilitate the development of more effective pipelines for protein-directed evolution.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.cels.2023.07.003>.

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AUTHOR CONTRIBUTIONS

Conceptualization, M.Z., C.L., and L.C.; methodology and software, L.C.; validation and formal analysis, L.C., Z.Z., and Z.L.; writing – original draft, L.C.; writing – review & editing, L.C., M.Z., Z.Z., Z.L., R.L., R.H., L.C., D.W., X.L., C.L., and K.C.; supervision, M.Z., C.L., and K.C.

DECLARATION OF INTERESTS

The authors declare no competing interests.

INCLUSION AND DIVERSITY

We support inclusive, diverse, and equitable conduct of research.

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REFERENCES

1. Arnold, F.H. (1998). Design by directed evolution. *Acc. Chem. Res.* **31**, 125–131.
2. Saito, Y., Oikawa, M., Nakazawa, H., Niide, T., Kameda, T., Tsuda, K., and Umetsu, M. (2018). Machine-learning-guided mutagenesis for directed evolution of fluorescent proteins. *ACS Synth. Biol.* **7**, 2014–2022. <https://doi.org/10.1021/acssynbio.8b00155>.
3. Romero, P.A., Krause, A., and Arnold, F.H. (2013). Navigating the protein fitness landscape with Gaussian processes. *Proc. Natl. Acad. Sci. USA* **110**, E193–E201. <https://doi.org/10.1073/pnas.1215251110>.
4. Lu, H., Diaz, D.J., Czarnecki, N.J., Zhu, C., Kim, W., Shroff, R., Acosta, D.J., Alexander, B.R., Cole, H.O., Zhang, Y., et al. (2022). Machine learning-aided engineering of hydrolases for PET depolymerization. *Nature* **604**, 662–667. <https://doi.org/10.1038/s41586-022-04599-z>.
5. de Visser, J.A., and Krug, J. (2014). Empirical fitness landscapes and the predictability of evolution. *Nat. Rev. Genet.* **15**, 480–490. <https://doi.org/10.1038/nrg3744>.
6. Diss, G., and Lehner, B. (2018). The genetic landscape of a physical interaction. *eLife* **7**, e32472. <https://doi.org/10.7554/eLife.32472>.
7. Gonzalez, C.E., and Ostermeier, M. (2019). Pervasive pairwise intragenic epistasis among sequential mutations in TEM-1 beta-lactamase. *J. Mol. Biol.* **431**, 1981–1992. <https://doi.org/10.1016/j.jmb.2019.03.020>.
8. Wu, N.C., Dai, L., Olson, C.A., Lloyd-Smith, J.O., and Sun, R. (2016). Adaptation in protein fitness landscapes is facilitated by indirect paths. *eLife* **5**, e16965. <https://doi.org/10.7554/eLife.16965>.
9. Sarkisyan, K.S., Bolotin, D.A., Meer, M.V., Usmanova, D.R., Mishin, A.S., Sharonov, G.V., Ivankov, D.N., Bozhanova, N.G., Baranov, M.S., Soylemez, O., et al. (2016). Local fitness landscape of the green fluorescent protein. *Nature* **533**, 397–401. <https://doi.org/10.1038/nature17995>.
10. Olson, C.A., Wu, N.C., and Sun, R. (2014). A comprehensive biophysical description of pairwise epistasis throughout an entire protein domain. *Curr. Biol.* **24**, 2643–2651. <https://doi.org/10.1016/j.cub.2014.09.072>.
11. Araya, C.L., Fowler, D.M., Chen, W., Muniez, I., Kelly, J.W., and Fields, S. (2012). A fundamental protein property, thermodynamic stability, revealed solely from large-scale measurements of protein function. *Proc. Natl. Acad. Sci. USA* **109**, 16858–16863. <https://doi.org/10.1073/pnas.1209751109>.
12. Sailer, Z.R., and Harms, M.J. (2017). High-order epistasis shapes evolutionary trajectories. *PLoS Comput. Biol.* **13**, e1005541. <https://doi.org/10.1371/journal.pcbi.1005541>.
13. Starr, T.N., and Thornton, J.W. (2016). Epistasis in protein evolution. *Protein Sci.* **25**, 1204–1218. <https://doi.org/10.1002/pro.2897>.
14. Domingo, J., Diss, G., and Lehner, B. (2018). Pairwise and higher-order genetic interactions during the evolution of a tRNA. *Nature* **558**, 117–121. <https://doi.org/10.1038/s41586-018-0170-7>.
15. Hopf, T.A., Ingraham, J.B., Poelwijk, F.J., Schärfe, C.P., Springer, M., Sander, C., and Marks, D.S. (2017). Mutation effects predicted from sequence co-variation. *Nat. Biotechnol.* **35**, 128–135. <https://doi.org/10.1038/nbt.3769>.
16. Riesselman, A.J., Ingraham, J.B., and Marks, D.S. (2018). Deep generative models of genetic variation capture the effects of mutations. *Nat. Methods* **15**, 816–822. <https://doi.org/10.1038/s41592-018-0138-4>.
17. Rives, A., Meier, J., Sercu, T., Goyal, S., Lin, Z., Liu, J., Guo, D., Ott, M., Zitnick, C.L., Ma, J., et al. (2021). Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences. *Proc. Natl. Acad. Sci. USA* **118**, e2016239118. <https://doi.org/10.1073/pnas.2016239118>.
18. Meier, J., Rao, R., Verkuil, R., Liu, J., Sercu, T., and Rives, A. (2021). Language models enable zero-shot prediction of the effects of mutations on protein function. *bioRxiv*. <https://doi.org/10.1101/2021.07.09.450648>.
19. Suzek, B.E., Wang, Y., Huang, H., McGarvey, P.B., and Wu, C.H.; UniProt Consortium (2015). UniRef clusters: a comprehensive and scalable alternative for improving sequence similarity searches. *Bioinformatics* **31**, 926–932. <https://doi.org/10.1093/bioinformatics/btu739>.
20. Rao, R., Liu, J., Verkuil, R., Meier, J., Canny, J.F., Abbeel, P., Sercu, T., and Rives, A. (2021). MSA transformer. Preprint at bioRxiv. <https://doi.org/10.1101/2021.02.12.430858>.
21. Luo, Y., Jiang, G., Yu, T., Liu, Y., Vo, L., Ding, H., Su, Y., Qian, W.W., Zhao, H., and Peng, J. (2021). ECNet is an evolutionary context-integrated deep learning framework for protein engineering. *Nat. Commun.* **12**, 5743. <https://doi.org/10.1038/s41467-021-25976-8>.
22. Rao, R. (2019). Evaluating Protein Transfer Learning with TAPE. *NeurIPS. Advances in Neural Information Processing Systems 32 (NeurIPS 2019)*.
23. Seemayer, S., Gruber, M., and Söding, J. (2014). CCMpred-fast and precise prediction of protein residue-residue contacts from correlated mutations. *Bioinformatics* **30**, 3128–3130. <https://doi.org/10.1093/bioinformatics/btu500>.
24. Gelman, S., Fahlberg, S.A., Heinzelman, P., Romero, P.A., and Gitter, A. (2021). Neural networks to learn protein sequence-function relationships from deep mutational scanning data. *Proc. Natl. Acad. Sci. USA* **118**, e2104878118. <https://doi.org/10.1073/pnas.2104878118>.
25. Riley, P. (2019). Three pitfalls to avoid in machine learning. *Nature* **572**, 27–29.
26. Hsu, C., Nisonoff, H., Fannjiang, C., and Listgarten, J. (2022). Learning protein fitness models from evolutionary and assay-labeled data. *Nat. Biotechnol.* **40**, 1114–1122. <https://doi.org/10.1038/s41587-021-01146-5>.

27. Dunham, A.S., and Beltrao, P. (2021). Exploring amino acid functions in a deep mutational landscape. *Mol. Syst. Biol.* 17, e10305. <https://doi.org/10.15252/msb.202110305>.
28. Gray, V.E., Hause, R.J., Luebeck, J., Shendure, J., and Fowler, D.M. (2018). Quantitative missense variant effect prediction using large-scale mutagenesis data. *Cell Syst.* 6, 116–124.e3. <https://doi.org/10.1016/j.cels.2017.11.003>.
29. Hoie, M.H., Cagiada, M., Beck Frederiksen, A.H., Stein, A., and Lindorff-Larsen, K. (2022). Predicting and interpreting large-scale mutagenesis data using analyses of protein stability and conservation. *Cell Rep.* 38, 110207. <https://doi.org/10.1016/j.celrep.2021.110207>.
30. Park, H., Bradley, P., Greisen, P., Jr., Liu, Y., Mulligan, V.K., Kim, D.E., Baker, D., and DiMaio, F. (2016). Simultaneous optimization of biomolecular energy functions on features from small molecules and macromolecules. *J. Chem. Theory Comput.* 12, 6201–6212. <https://doi.org/10.1021/acs.jctc.6b00819>.
31. Frenz, B., Lewis, S.M., King, I., DiMaio, F., Park, H., and Song, Y. (2020). Prediction of protein mutational free energy: benchmark and sampling improvements increase classification accuracy. *Front. Bioeng. Biotechnol.* 8, 558247. <https://doi.org/10.3389/fbioe.2020.558247>.
32. Leman, J.K., Weitzner, B.D., Lewis, S.M., Adolf-Bryfogle, J., Alam, N., Alford, R.F., Aprahamian, M., Baker, D., Barlow, K.A., Barth, P., et al. (2020). Macromolecular modeling and design in Rosetta: recent methods and frameworks. *Nat. Methods* 17, 665–680. <https://doi.org/10.1038/s41592-020-0848-2>.
33. Rollins, N.J., Brock, K.P., Poelwijk, F.J., Stiffler, M.A., Gauthier, N.P., Sander, C., and Marks, D.S. (2019). Inferring protein 3D structure from deep mutation scans. *Nat. Genet.* 51, 1170–1176. <https://doi.org/10.1038/s41588-019-0432-9>.
34. Schmiedel, J.M., and Lehner, B. (2019). Determining protein structures using deep mutagenesis. *Nat. Genet.* 51, 1177–1186. <https://doi.org/10.1038/s41588-019-0431-x>.
35. Sim, N.L., Kumar, P., Hu, J., Henikoff, S., Schneider, G., and Ng, P.C. (2012). SIFT web server: predicting effects of amino acid substitutions on proteins. *Nucleic Acids Res.* 40, W452–W457. <https://doi.org/10.1093/nar/gks539>.
36. Podgornaia, A.I., and Laub, M.T. (2015). Protein evolution. Pervasive degeneracy and epistasis in a protein-protein interface. *Science* 347, 673–677. <https://doi.org/10.1126/science.1257360>.
37. Adzhubei, I.A., Schmidt, S., Peshkin, L., Ramensky, V.E., Gerasimova, A., Bork, P., Kondrashov, A.S., and Sunyaev, S.R. (2010). A method and server for predicting damaging missense mutations. *Nat. Methods* 7, 248–249. <https://doi.org/10.1038/nmeth0410-248>.
38. Shihab, H.A., Gough, J., Cooper, D.N., Stenson, P.D., Barker, G.L., Edwards, K.J., Day, I.N., and Gaunt, T.R. (2013). Predicting the functional, molecular, and phenotypic consequences of amino acid substitutions using hidden Markov models. *Hum. Mutat.* 34, 57–65. <https://doi.org/10.1002/humu.22225>.
39. Jumper, J., Evans, R., Pritzel, A., Green, T., Figurnov, M., Ronneberger, O., Tunyasuvunakool, K., Bates, R., Židek, A., Potapenko, A., et al. (2021). Highly accurate protein structure prediction with AlphaFold. *Nature* 596, 583–589. <https://doi.org/10.1038/s41586-021-03819-2>.
40. Gligorijević, V., Renfrew, P.D., Kosciółek, T., Leman, J.K., Berenberg, D., Vatanen, T., Chandler, C., Taylor, B.C., Fisk, I.M., Vlamakis, H., et al. (2021). Structure-based protein function prediction using graph convolutional networks. *Nat. Commun.* 12, 3168. <https://doi.org/10.1038/s41467-021-23303-9>.
41. Baek, M., DiMaio, F., Anishchenko, I., Dauparas, J., Ovchinnikov, S., Lee, G.R., Wang, J., Cong, Q., Kinch, L.N., Schaeffer, R.D., et al. (2021). Accurate prediction of protein structures and interactions using a three-track neural network. *Science* 373, 871–876.
42. Chen, C., Chen, X., Morehead, A., Wu, T., and Cheng, J. (2023). 3D-equivariant graph neural networks for protein model quality assessment. *Bioinformatics* 39, btad030. <https://doi.org/10.1101/2022.04.12.488060>.
43. Hsu, C., Verkuil, R., Liu, J., Lin, Z., Hie, B., Sercu, T., Lerer, A., and Rives, A. (2022). Learning inverse folding from millions of predicted structures. Preprint at bioRxiv. <https://doi.org/10.1101/2022.04.10.487779>.
44. Jing, B., Eismann, S., Suriana, P., Townshend, R.J.L., and Dror, R. (2021). Learning from protein structure with geometric vector perceptrons. Preprint at arXiv. <https://doi.org/10.48550/arXiv.2009.01411>.
45. Firnberg, E., Labonte, J.W., Gray, J.J., and Ostermeier, M. (2014). A comprehensive, high-resolution map of a gene's fitness landscape. *Mol. Biol. Evol.* 31, 1581–1592. <https://doi.org/10.1093/molbev/msu081>.
46. McLaughlin, R.N., Jr., Poelwijk, F.J., Raman, A., Gosal, W.S., and Ranganathan, R. (2012). The spatial architecture of protein function and adaptation. *Nature* 491, 138–142. <https://doi.org/10.1038/nature11500>.
47. Shin, J.E., Riesselman, A.J., Kollasch, A.W., McMahon, C., Simon, E., Sander, C., Manglik, A., Kruse, A.C., and Marks, D.S. (2021). Protein design and variant prediction using autoregressive generative models. *Nat. Commun.* 12, 2403. <https://doi.org/10.1038/s41467-021-22732-w>.
48. Yang, K.K., Wu, Z., Bedbrook, C.N., and Arnold, F.H. (2018). Learned protein embeddings for machine learning. *Bioinformatics* 34, 2642–2648. <https://doi.org/10.1093/bioinformatics/bty178>.
49. Alley, E.C., Khimulya, G., Biswas, S., AlQuraishi, M., and Church, G.M. (2019). Unified rational protein engineering with sequence-based deep representation learning. *Nat. Methods* 16, 1315–1322. <https://doi.org/10.1038/s41592-019-0598-1>.
50. Biswas, S., Khimulya, G., Alley, E.C., Esvelt, K.M., and Church, G.M. (2021). Low-N protein engineering with data-efficient deep learning. *Nat. Methods* 18, 389–396. <https://doi.org/10.1038/s41592-021-01100-y>.
51. Barrat-Charlaix, P., Figliuzzi, M., and Weigt, M. (2016). Improving landscape inference by integrating heterogeneous data in the inverse Ising problem. *Sci. Rep.* 6, 37812. <https://doi.org/10.1038/srep37812>.
52. Shamsi, Z., Chan, M., and Shukla, D. (2020). TLMutation: predicting the effects of mutations using transfer learning. *J. Phys. Chem. B* 124, 3845–3854. <https://doi.org/10.1021/acs.jpcc.0c00197>.
53. Wittmann, B.J., Yue, Y., and Arnold, F.H. (2021). Informed training set design enables efficient machine learning-assisted directed protein evolution. *Cell Syst.* 12, 1026–1045.e7. <https://doi.org/10.1016/j.cels.2021.07.008>.
54. Gonzalez Somermeyer, L., Fleiss, A., Mishin, A.S., Bozhanova, N.G., Igoikina, A.A., Meiler, J., Alaball Pujol, M.E., Putintseva, E.V., Sarkisyan, K.S., and Kondrashov, F.A. (2022). Heterogeneity of the GFP fitness landscape and data-driven protein design. *eLife* 11, e75842. <https://doi.org/10.7554/eLife.75842>.
55. Weinreich, D.M., Delaney, N.F., Depristo, M.A., and Hartl, D.L. (2006). Darwinian evolution can follow only very few mutational paths to fitter proteins. *Science* 312, 111–114.
56. Tufts, D.M., Natarajan, C., Revsbech, I.G., Projecto-Garcia, J., Hoffmann, F.G., Weber, R.E., Fago, A., Moriyama, H., and Storz, J.F. (2015). Epistasis constrains mutational pathways of hemoglobin adaptation in high-altitude pikas. *Mol. Biol. Evol.* 32, 287–298. <https://doi.org/10.1093/molbev/msu311>.
57. Starr, T.N., Greaney, A.J., Hilton, S.K., Ellis, D., Crawford, K.H.D., Dingens, A.S., Navarro, M.J., Bowen, J.E., Tortorici, M.A., Walls, A.C., et al. (2020). Deep mutational scanning of SARS-CoV-2 receptor binding domain reveals constraints on folding and ACE2 binding. *Cell* 182, 1295–1310.e20. <https://doi.org/10.1016/j.cell.2020.08.012>.
58. Taft, J.M., Weber, C.R., Gao, B., Ehling, R.A., Han, J., Frei, L., Metcalfe, S.W., Overath, M.D., Yermanos, A., Kelton, W., et al. (2022). Deep mutational learning predicts ACE2 binding and antibody escape to combinatorial mutations in the SARS-CoV-2 receptor-binding domain. *Cell* 185, 4008–4022.e14. <https://doi.org/10.1016/j.cell.2022.08.024>.
59. Sandberg, W.S., and Terwilliger, T.C. (1993). Engineering multiple properties of a protein by combinatorial mutagenesis. *Proc. Natl. Acad. Sci. USA* 90, 8367–8371.
60. Bershtein, S., Segal, M., Bekerman, R., Tokuriki, N., and Tawfik, D.S. (2006). Robustness-epistasis link shapes the fitness landscape of a

- p randomly drifting protein.
- Nature*
- 444, 929–932.
- <https://doi.org/10.1038/nature05385>
- .
61. Wang, Z. (2020). Gradient vaccine: investigating and improving multi-task optimization in massively multilingual models. Preprint at arXiv. <https://doi.org/10.48550/arXiv.2010.05874>.
 62. Yu, T. (2020). Gradient surgery for multi-task learning. *NeurIPS. Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*.
 63. Stiffler, M.A., Hekstra, D.R., and Ranganathan, R. (2015). Evolvability as a function of purifying selection in TEM-1 beta-lactamase. *Cell* 160, 882–892. <https://doi.org/10.1016/j.cell.2015.01.035>.
 64. Chen, J.Z., Fowler, D.M., and Tokuriki, N. (2022). Environmental selection and epistasis in an empirical phenotype-environment-fitness landscape. *Nat. Ecol. Evol.* 6, 427–438. <https://doi.org/10.1038/s41559-022-01675-5>.
 65. Ding, X., Cui, R., Yu, J., Liu, T., Zhu, T., Wang, D., Chang, J., Fan, Z., Liu, X., Chen, K., et al. (2021). Active learning for drug design: A case study on the plasma exposure of orally administered drugs. *J. Med. Chem.* 64, 16838–16853. <https://doi.org/10.1021/acs.jmedchem.1c01683>.
 66. Yu, J., Li, X., and Zheng, M. (2021). Current status of active learning for drug discovery. *Artif. Intell. Life Sci.* 1, 100023. <https://doi.org/10.1016/j.ailsci.2021.100023>.
 67. Hopf, T.A., Green, A.G., Schubert, B., Mersmann, S., Schärfe, C.P.I., Ingraham, J.B., Toth-Petroczy, A., Brock, K., Riesselman, A.J., Palmedo, P., et al. (2019). The EVcouplings Python framework for coevolutionary sequence analysis. *Bioinformatics* 35, 1582–1584. <https://doi.org/10.1093/bioinformatics/bty862>.
 68. Eddy, S.R. (2011). Accelerated profile HMM Searches. *PLoS Comput. Biol.* 7, e1002195. <https://doi.org/10.1371/journal.pcbi.1002195>.
 69. Vaswani, A. (2017). Attention is all you need. 31st Conference on Neural Information Processing Systems. *Advances in Neural Information Processing Systems 30 (NIPS 2017)*.
 70. Srivastava, N. (2014). Dropout: A simple way to prevent neural networks from overfitting. *J. Mach. Learn. Res.* 15, 1929–1958.
 71. Khan, A.I., Dinh, D.M., Schneider, D., Lenski, R.E., and Cooper, T.F. (2011). Negative epistasis between beneficial mutations in an evolving bacterial population. *Science* 332, 1193–1196. <https://doi.org/10.1126/science.1203801>.

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited Data		
Processed DMS data files; pretrained model parameters	This study	Zenodo: https://doi.org/10.5281/zenodo.8103840 .
Software and algorithms		
Custom code	This study	https://github.com/cl6666666/GVP-MSA . Zenodo: https://doi.org/10.5281/zenodo.8146299 .
Python	https://www.python.org/	v3.9
Rosetta	Leman et al. ³²	Rosetta 2022.11 released on March 13, 2022
AlphaFold2	Jumper et al. ³⁹	https://github.com/deepmind/alphafold
DeepSequence	Riesselman et al. ¹⁶	https://github.com/debbiemarkslab/DeepSequence
EVmutation	Hopf et al. ¹⁵	https://github.com/debbiemarkslab/EVcouplings
ECnet	Luo et al. ²¹	https://github.com/luoyunan/ECNet
Evolutionary Scale Modeling (ESM)	Rives et al. ¹⁷ ; Rao et al. ²⁰	https://github.com/facebookresearch/esm
Augmented DeepSequence	Hsu et al. ^{26,43}	https://github.com/chloechsu/combining-evolutionary-and-assay-labelled-data

RESOURCE AVAILABILITY

Lead Contact

Further information and requests for resources should be directed to and will be fulfilled by the Lead Contact, Mingyue Zheng (myzheng@simm.ac.cn).

Materials Availability

This study did not generate new materials.

Data and Code availability

- Data needed to replicate simulations have been deposited at Zenodo and are publicly available as of the date of publication. DOIs are listed in the [key resources table](#). This paper analyzes existing, publicly available DMS datasets. Detailed information of all DMS datasets used in this study are available in [Table S1](#).
- All original code has been deposited at <https://github.com/cl6666666/GVP-MSA> and is publicly available as of the date of publication. DOIs are listed in the [key resources table](#).
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

METHOD DETAILS

Datasets and preprocessing

Dataset collection

We have collected a broad range of mutation effect data sets generated by mutagenesis experiments of previous studies. First, diverse DMS datasets curated by Riesselman et al.¹⁶ and Gray et al.²⁸ were collected. We excluded studies for RNAs, influenza virus sequences, and datasets with fewer than 200 labeled fitness data. Finally, 36 mutagenesis datasets were obtained for benchmarking protein fitness landscapes modeling of single variants. To test the model's ability to predict higher-order variant effects, DMS datasets with both single variants and higher-order variants were also collected, including YAP1 WW,¹¹ TEM1,⁷ FOSJUN,⁶ avGFP,⁹ GB1 (2combo),¹⁰ and GB1 (4combo).⁸ These datasets contain a broad range of protein functions and their fitness was measured based on

different metrics such as growth rate, fluorescence brightness, and enzyme activity. When no experimental determined protein structure was available, the protein structures were predicted by AlphaFold2 using the default parameters and pipelines provided on the GitHub repository (<https://github.com/deepmind/alphafold>). A list of the datasets is shown in Table S1.

Generation of multiple sequence alignment (MSA)

MSA was generated using the EVcouplings web server (<https://v1.evcouplings.org/job>).⁶⁷ We used the default parameters of EVcouplings and followed the MSA generation protocol described in EVmutation.¹⁵ The homology search tool Jackhmmer⁶⁸ was used to search with five search iterations against the protein sequences database UniRef100.¹⁹ The bit core of 0.5 bits/residue was applied as the inclusion threshold for candidate sequences during alignment. We gradually lowered the bit core threshold by 0.1 bits per residue if there were not enough sequences (the number of effective sequences < 10L, where L is the length of the aligned sequence). The number of effective sequences in the alignment was calculated after down-weighting redundant sequences with a 0.8 similarity threshold.

Normalization of fitness among different DMS datasets

The fitness of each variant was measured by the relative frequency of sequences before and after experiment selection. Since different DMS experiments had their methods to calculate the fitness of variants, we need to do normalizations across different DMS datasets to make them comparable. First, if the fitness (F_a) was not in logarithm form, we redefined it as the natural logarithm of the ratio of the score of the variant (W_a) versus the wild-type score (W_{wt}), that is,

$$F_a = \ln\left(\frac{W_a}{W_{wt}}\right) \quad (\text{Equation 1})$$

Where the score W is the sequence count before or after selection. If W_a , that is the sequence count after selection, is zero, we would discard this sample. Mutations that show neutral effects have zero values of F_a , and those with advantageous or deleterious effects have a positive or negative value of F_a . Then, we normalized the fitness by dividing the absolute value of the lowest 5% of the fitness score, which is expressed as

$$F_{\text{normalized } i} = F_i / |F_{\text{bottom 5\%}}| \quad (\text{Equation 2})$$

We attempted to normalize the fitness data by leveraging the assumption that mutations that result in a complete loss of function are comparable among different DMS datasets.

Model architecture

We built a deep learning model to learn the fitness landscape, that is the mapping from sequence variants to fitness. The model architectures have two main modules that take as input the MSA and structure of the target protein, respectively. The MSA information was extracted by the pre-trained protein language model MSA Transformer and its parameters were fixed during all the training. The first row of MSA embedding in the last layer of the MSA Transformer was taken as the output MSA representation, and this MSA representation was projected using a linear fully connected layer to reduce its feature dimension, which was then concatenated into the final protein graph's nodes representation. For the protein structure encoding, we incorporated only the backbone of the protein structure in the graph neural network, as precise coordination of sidechains cannot be obtained when the protein structure is unavailable and required to be predicted by AlphaFold2. The architectures dealing with protein structure were based on GVP-GNN from Jing. et al.⁴⁴ We encoded each protein's wild-type structure as a graph $G = (V; E)$, where each node V represented features of residues and edge E represented the relationship between residues. We included 20 nearest neighbors of each node as edges which was measured by the distance between their $C\alpha$ atoms. Then, both scalar and vector features of nodes and edges were encoded. Scalar features of nodes included amino acid identity encoded by one-hot encoded representation, the first row of the MSA representation from the previously described MSA embedding module, and dihedral angles calculated from backbone atoms. Vector features of nodes included the imputed direction from $C\alpha$ to $C\beta$ atom and the directions of $C\alpha$ from this residue to its previous and next residues. Edge features included the direction of $C\alpha$ atoms between nodes, the spatial distance in the protein structure, and the topological distance in the primary sequences. At the graph propagation step, two layers of GVP-GNN were performed to update node embeddings from features of neighboring nodes and edges. More details on graph construction and message propagation can be found in the original report of Jing. et al. Then, the scalar feature of nodes in the last GVP-GNN layer was extracted as the representation of the protein graph and passed into a top module with multiple layers. The top module first contained a weighted averaged approach to project sequences with different lengths into one dimension. The weight is learned from the node representations in a self-attention⁶⁹ way, thus allowing the model to learn to pay more attention to residues around the mutation site or other important residues. The output is then transformed by a batch normalization layer and a fully-connected layer with tanh activation. Finally, a fully connected layer is used to map the hidden representation from the previous layer to obtain the predicted fitness. When training this model in a multi-task way, the parameters of the top module were task-specific while other parameters were shared across different mutagenesis datasets. Also, dropout layers were applied with a 10% dropout probability to prevent overfitting.⁷⁰

Training details

Protein-specific modeling

First, we validated our model's capability in learning the fitness landscape of a specific protein. A specific deep learning model will be trained for each protein mutagenesis dataset. We benchmarked each method using two dataset splitting strategies referred to as

random split and position-specific split, respectively. For each splitting strategy, 20% of the data set was sampled as test data, and the remaining 80% of the dataset is further split into 7/8 of the training set and 1/8 of the validation set. Training data was used to train the model to derive a specific set of parameters, and the validation set was used to select the hyperparameters and determine how many epochs to reach training convergence. Different from the random sample train/validation/test dataset of the random split strategy, the position-specific split strategy divided the mutational positions into train/validation/test based on their mutational position. The position-specific split strategy makes sure that the mutational position will not overlap between split datasets thus evaluating the performance of positional extrapolation. The hyperparameters to be selected included the number of GVP-GNN layers $L=2$ and 3, the maximum number of neighbors considered in protein graph construction $K=10, 15, 20, 25$, and 30, the projected embedding dimension of the first row of MSA representation $D=128$ and 256, and the learning rate of the Adam optimizer $lr=1e-4, 5e-5$ and $1e-5$. We performed a grid search for these hyperparameters on three randomly selected DMS datasets BG_STRSQ, BLAT_ECOLDX_Ranganathan2015, and DLG4 (detailed information of DMS dataset in Table S1). We found that the models with hyperparameters $L=2, K=15, D=128$, and $lr=1e-4$ showed better results. We did not search hyperparameters extensively for each protein DMS dataset due to the large computational costs, even though this will improve model performance. Since we regard it as a regression problem, the mean squared error between experimentally measured fitness and predicted fitness was used as the loss function. The max epoch running was set to 1500, and we used the early stop scheme with a patience value of 200 epochs, which means the model will stop training if the performance in the validation set has not improved for 200 epochs. At last, we trained the model with three replications of the same train/validation/test dataset using different training seeds and averaged these three replications to obtain ensembled predicted results. All these training and testing procedures were performed on one NVIDIA Tesla V100 (32G) GPU. Depending on different protein sequence lengths and data point sizes, it will take 1-20h to finish the training and testing procedures of one protein mutagenesis dataset.

Multi-task framework facilitated Protein-specific modeling

In this training scenario, we tested whether the leverage of other protein fitness data can facilitate the modeling of a specific protein. First, we excluded datasets that measured the binding ability to DNA or RNA, as well as datasets that measured the fitness of the same protein sequences as test proteins to avoid data leakage. Then, several DMS datasets were used to train a reference multi-task model. As it was shown in Figure S1A, the performance of nine datasets were slightly better than others. So, nine protein DMS datasets with higher performance in protein-specific training scenario were selected to train the reference model. Then, the reference model was used to finetune the fitness data of target protein. In the multi-task framework, all the parameters were shared among different protein datasets except that the top module was protein-specific. All the parameters can be changed in the finetuning procedure. The learning rate of the Adam optimizer was $5e-5$ for reference model training and $1e-5$ for finetuning on new datasets. The max epoch running was set to 800 and the patience value was 100 epochs for reference model training.

Zero-shot predictor

In this application scenario, we train a model using other proteins' fitness landscape data to predict a new protein's fitness. We selected several DMS datasets containing variants with multiple substitutions as study datasets: YAP1 WW,¹¹ TEM1,⁷ FOSJUN,⁶ avGFP,⁹ GB1 (2combo),¹⁰ and GB1 (4combo).⁸ We also included as the training set six DMS datasets of single variants with higher performance in protein-specific training scenario, including B3VI55_LIPSTSTABLE, BG_STRSQ, PTEN, AMIE_ACET, HSP90, and KKA2 (detailed information of DMS dataset in Table S1). The number of training dataset determination was shown in Figure S1B. We performed normalization (detailed procedures in the data normalization section) for every dataset to try to make them comparable, and employed a leave-one-protein-out training approach. Since GB1(2combo) and GB1(4combo) include different variant combinations of the same protein sequences, we need to remove one dataset from the training dataset to avoid data leakage. For example, if we tested the model's performance on dataset GB1(2combo), other datasets excluding GB1(4combo) would be used as training and validation sets. Also, we only retained one GB1 DMS dataset as the training or validation set to avoid the overrepresentation of the GB1 sequences. For example, when we tested the DMS dataset YAP1 WW with single and double variants, we would include only GB1(2combo) dataset and excluded GB1(4combo) dataset. Thus, when testing on each DMS dataset, there would be ten DMS datasets of different proteins as training resources. We randomly selected one dataset as the validation set to indicate when to stop the training procedure and ten different training procedures were performed. Then, these ten different training results were ensembled by simply averaging over different folds.

During this training procedure, we need to balance the number of training samples from different DMS datasets to avoid overrepresenting the properties of one specific protein with the larger dataset size. Therefore, we randomly selected training samples from different datasets and stopped the training process if all the data points from one dataset had been used up. In this way, the number of training data used from each dataset is equal in expectation and every variant sample in specific DMS datasets had the same chance to be selected to optimize the parameters of the model. Additionally, besides the regression object of the mean squared error loss, we applied classification loss to enhance the model's ability to prioritize the high-fitness variants. We also penalized the cross entropy loss on the class of advantageous variants with a weight value of 100 to deal with the unbalanced training set of advantageous variants and deleterious variants. The max epoch running was set to 80, and we used the early stop scheme with a patience value of 15 epochs. All these training and testing procedures were performed on one NVIDIA Tesla V100 (32G) GPU. Depending on different protein sequence lengths and data point sizes, it will take 10-15h to finish the training and testing procedures of one protein DMS dataset.

Extrapolation to higher-order mutation effects from single mutation effects

In this training scenario, we tested the capability of extrapolating to higher-order mutation effects when only single mutation effects were available. We selected several DMS datasets containing variants with multiple substitutions as study cases: YAP1 WW,¹¹ TEM1,⁷ FOSJUN,⁶ avGFP,⁹ GB1 (2combo),¹⁰ GB1 (4combo),⁸ and SARS-CoV-2.^{57,58} We trained GVP-MSA for extrapolation from single variants to higher-order variants by two strategies referred to as “Protein-specific” and “Multi-protein”, respectively.

For train/validation/test data split in “Protein-specific” training strategy, higher-order variants were selected as held-out test data, and the remaining single mutation samples were divided into 9/10 for the training set and 1/10 for the validation set. No samples with higher-order variants were used for model parameter training. As for “Multi-protein” strategy, we attempted to leverage higher-order variant effects of other reference DMS datasets to help the modeling of target protein without higher-order variant effect. For reference datasets, we split both single and higher-order variants into train/validation/test datasets with a ratio of 7:1:2. For the target DMS dataset, we applied the same data split procedure as in “Protein-specific” training strategy, and ensured that no higher-order variant effects in the train or validation set.

During this training procedure, we balance the number of training samples from different DMS datasets to avoid overrepresenting one specific protein in the same way as in zero-shot predictor training procedure. Moreover, since the training dataset of the target protein only includes single variants, we oversampled the target protein training dataset to include about 20000 samples to make sure the training sample number among different datasets was in the same multitude. To make full use of the single variant samples for training, we randomly split the train/validation set with ten different seeds, and ensembled these results through averaging. “Protein-specific”, “Multi-protein”, and ECNet were all evaluated using this ensemble strategy.

Calculation of epistasis value and identification of epistatic interactions

We calculated the epistasis (ϵ) using the absolute epistatic deviation model described by Kahn et al.⁷¹ The epistasis of a higher-order variant was defined as the difference between its fitness and the addition of the constituent single mutations’ fitness, that is,

$$\epsilon_M = W_M - \left(\sum_{i \in M} W_i \right)_{\text{capped}} \quad (\text{Equation 3})$$

Where M is the set of mutations, W_M is the fitness of the higher-order variant, and W_i is the fitness of the constituent single variants. Also, we confined the added value to the maximum or minimum fitness value measured in an assay when it exceeds the range of assayed fitness values. Variants with the most extreme epistasis values above 10% or below were classified as positive or negative epistatic variants. Due to the lack of a golden standard of epistasis effects, we identified positive or negative epistatic interactions based on the distribution of the epistasis value of a specific protein.

Rosetta $\Delta\Delta G$ predictions

We predicted the relative Gibbs free energy change ($\Delta\Delta G$) using the Rosetta 2022.11 released on March 13, 2022. We followed the Cartesian $\Delta\Delta G$ ^{30,31} protocols described in the documentation of Rosetta’s website (<https://www.rosettacommons.org/docs/latest/cartesian-ddg>). First, we relaxed the protein structure using the relax application. Then, the “cartesian_ddg” function was applied to predict the mutation effects. We iteratively predicted $\Delta\Delta G$ three times and calculated the average as the predicted score.

Calculation of the NDCG measure

NDCG is a commonly used measure of ranking quality in information retrieval. It was applied in measuring the model’s ability of assisting directed evolution recently as they emphasize the ability to identify high-performing data points.^{21,26,53} To derive NDCG, we first calculate the DCG of the predicted results and then normalize it by dividing it by the best DCG in the ideal ranking situation. The calculation of DCG is a weighted sum of true variant fitness by the logarithmic discount of the predicted rank order, that is:

$$DCG = \sum_{i=1}^N \frac{f_i}{\log_2(i+1)} \quad (\text{Equation 4})$$

Where f_i is the true fitness of the variant and i corresponds to the predicted rank of the variant. This equation will obtain a higher value if a high-fitness variant is higher ranked and the variants with low fitness will contribute less than those with high fitness. At last, we obtain NDCG by dividing it by the best DCG, that is,

$$NDCG = \frac{DCG_i}{DCG_{\text{best}}} \quad (\text{Equation 5})$$

The NDCG value ranges from 0 to 1 and a higher value of NDCG means a better ranking model. To ensure that the NDCG value is not negative, the fitness values are normalized to be positive by adding the absolute value of minimum fitness. This is only done if the minimum fitness value is negative. Moreover, we only consider the predicted performance of the top 1% fitness to avoid the NDCG being swamped by large number of variants with low fitness.

Using MAE to evaluate the ranking ability of a subgroup out of the entire population

We use MAE (mean absolute error) to evaluate the ability of models to rank a certain small subgroup out of the entire population. This is done by calculating the difference between the predicted rank and the experimental assayed rank. Here, we provide an example to illustrate how to calculate MAE and why we use it.

Assume that the experimental labeled data sorts 10 variants as 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, and the first 3 variants belong to the subgroup of positive epistasis. Two methods predict the ranking to be 2, 3, 4, 5, 6, 7, 8, 9, 10, 1 and 8, 9, 10, 1, 2, 3, 4, 5, 6, 7. We aim to evaluate if the model can select the positive epistasis subgroup in the context of the whole group. Therefore, we calculate the MAE of the first three ranks of the predicted and experimental labeled data, and then divide it by the total number of the variants. In this case, the calculated MAE metrics for these two ranking will be 0.1 and 0.7, indicating that the first method performs better than the second. This is concordant with the superior ability of the first method. However, if we calculate the Spearman's correlation coefficients or other ranking metrics of the first three variants, it will obtain the same results between these two methods.

Baseline methods

For supervised training, we compared our method GVP-MSA to ECNet, linear model from Hsu et al.,²⁶ and augmented DeepSequence from Hsu et al. For unsupervised training, we compared our method to zero-shot predictors EVmutation, DeepSequence, ESM-1b, ESM-1v, and ESM-MSA-1b. We also devised simple baseline models based on the intrinsic tendency of the fitness data. For predicting the fitness of single variants when splitting datasets randomly, we designed the Average model. For extrapolating to higher-order variant fitness from single variants, we designed the Addition model.

Average model

The predicted fitness of a variant equals the average fitness of mutations occurring at the same positions as the tested variant. When calculating the average value, the highest and the lowest fitness were abandoned if this position had more than two fitness data points in the training data set. If there is no known fitness of variants at that position, we will fill the position with the average fitness of all variants.

Addition model

The predicted fitness of higher-order variants equals the addition of their constituent single variants. If there is no known experimental fitness value of the single variant, we will predict the fitness of single variants by the above Average model.

ECNet

ECNet combines the evolutionary contexts from MSA and global contexts from the protein sequences pre-training model to learn the variant effects. We compared our model GVP-MSA to ECNet in two training scenarios: (1) random or positional extrapolation of single variants and (2) higher-order variants extrapolation from single variants. The model was executed using the parameters described in the original work.²¹ The train/validation/test datasets were split using the same seeds when compared with GVP-MSA.

Linear model and Augmented DeepSequence

The linear model takes the one-hot encoded sequence representation as input and applies L2 regularization for the ridge regression. The Augmented DeepSequence applies the ridge regression to combine the probability density feature from DeepSequence and the experimental assayed fitness data. The model was executed using the parameters described in the original work²⁶ and the results of DeepSequence were extracted directly from their published predicted results. The train/validation/test datasets were split using the same seeds when compared with GVP-MSA.

EVmutation and DeepSequence

EVmutation and DeepSequence predict the variant effects by learning the latent distribution of multiple sequence alignment (MSA) of the pairwise or higher-order dependencies. EVmutation was executed using the example code provided on the EVcouplings GitHub repository (<https://github.com/debbiemarkslab/EVcouplings>). For DeepSequence calculation, we also followed the example code provided on the DeepSequence GitHub repository (<https://github.com/debbiemarkslab/DeepSequence>). We first trained a VAE model to the sequence alignments and calculated the mutation effect prediction using 2,000 prediction iterations.

ESM-1b, ESM-1v, and ESM-MSA-1b

Protein language models can predict the variant effects by calculating the logits of the masked mutation positions when passing the masked sequences or alignments into the model. For higher-order variants, we used the product of the probability of their constituent single variants as the predicted score, which was shown to achieve similar performance as the conditional probability protocols described in Wittmann et al.⁵³ To perform the calculations, we used the code provided in Wittmann et al. (<https://github.com/fhalab/MLDE>).

Supplemental information

**Learning protein fitness landscapes with deep
mutational scanning data from multiple sources**

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Supplementary Information

Supplementary Figures

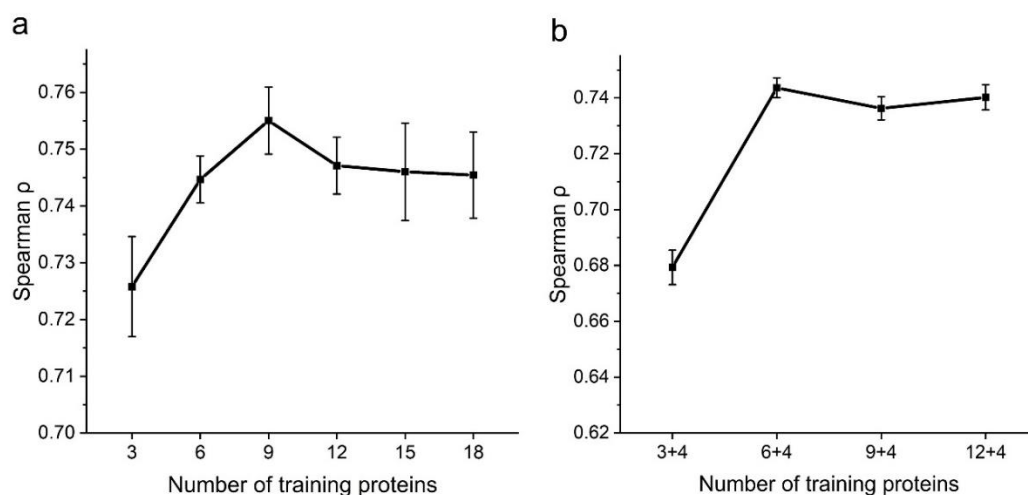


Figure S1. Analysis of the number of protein datasets used to train the multi-protein model. We gradually added more DMS datasets to train the multi-protein model by selecting datasets with better performance in the protein-specific training scenario. **a**, Performance was tested in the scenario of multi-protein training that facilitate protein-specific modeling. We tested the model's performance in both random and position-specific splitting scenarios on three DMS datasets: BG_STRSQ, BLAT_ECOLX_Ranganathan2015, and DLG4. The y-axis represents the mean Spearman's correlation coefficient across different test systems. **b**, Performance was tested in the scenario of zero-shot predictions. The "3+4" means three DMS datasets with only single variants and four DMS datasets with higher-order variants were used to train the zero-shot predictors. The y-axis represents the performance of the test DMS dataset TEM1.

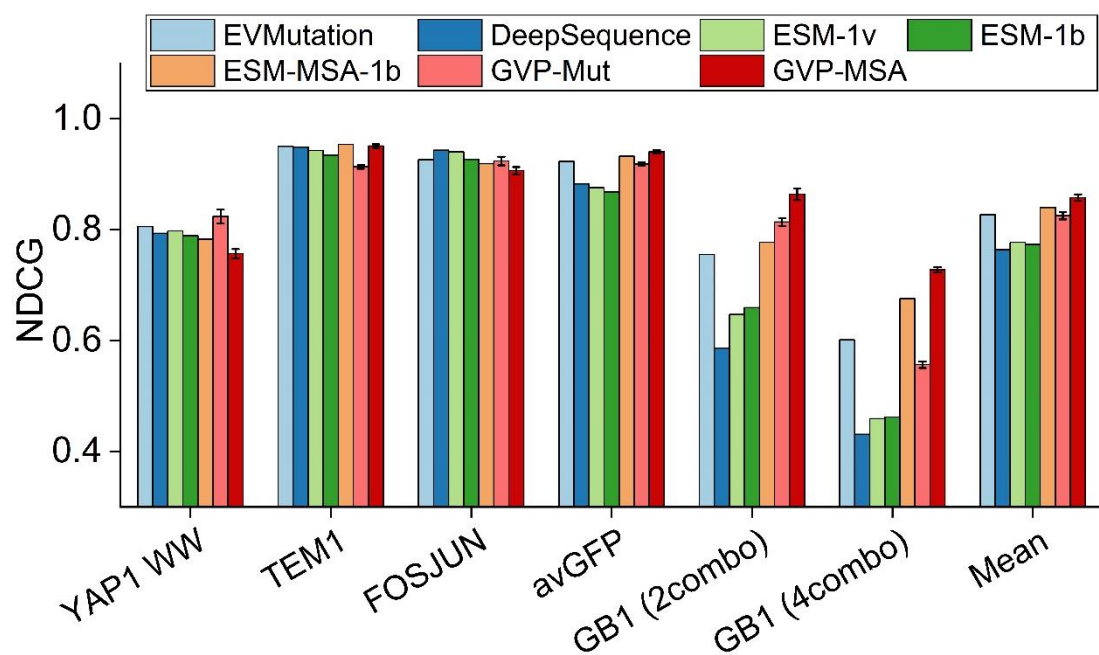


Figure S2. The NDCG performance of unsupervised GVP-MSA. Comparisons to five unsupervised models on six different DMS datasets. The ‘Mean’ column is the mean NDCG value among six datasets.

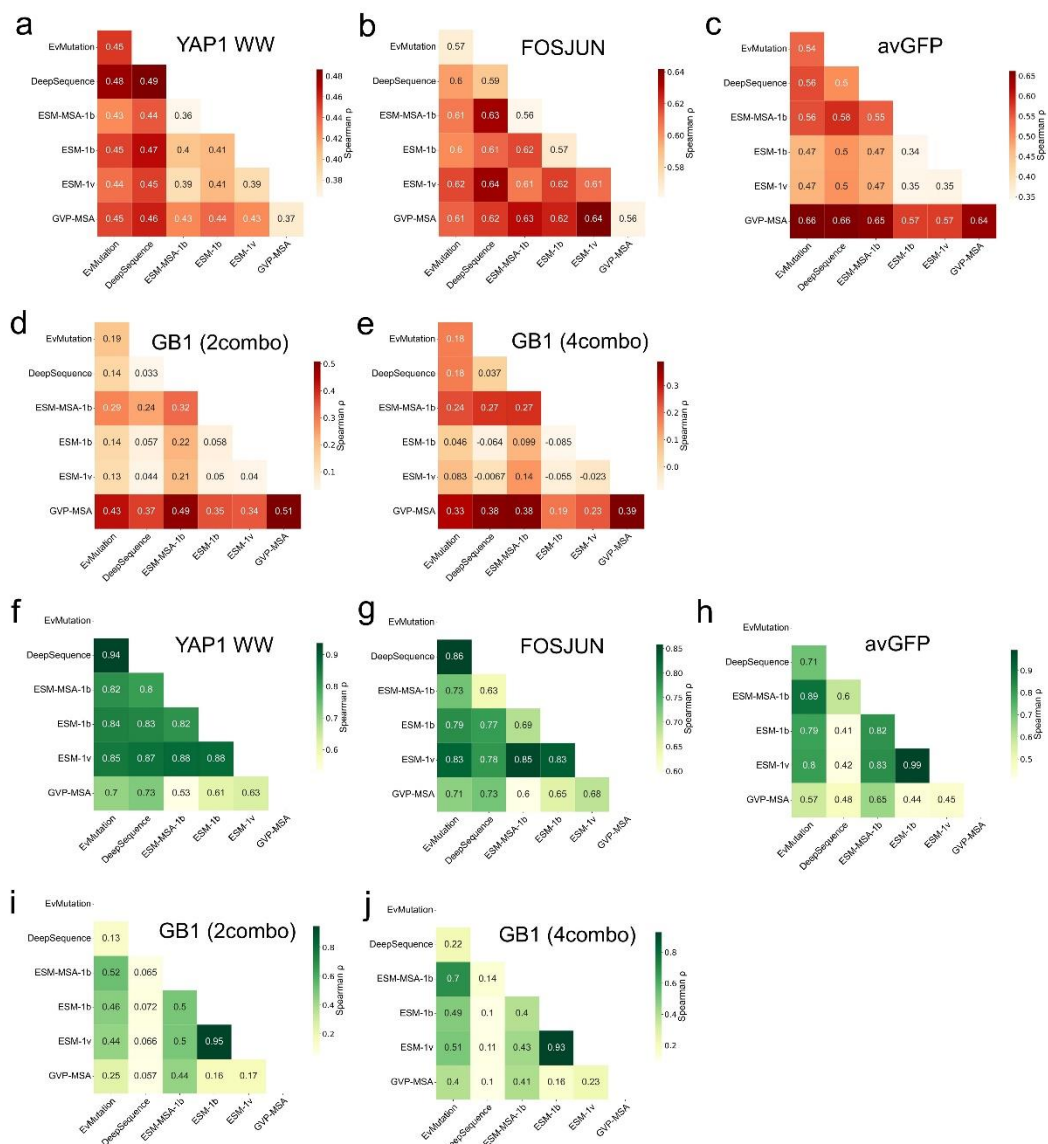


Figure S3. The performance of pair-wise ensembled methods and the pair-wise Spearman correlation. a-e, The performance of pair-wise ensembled methods of different DMS datasets. The diagonal value shows the Spearman's correlation coefficient of the predicted fitness with experimental fitness from a single method. **f-j,** The pair-wise Spearman's correlation coefficient of the predicted fitness from different methods.

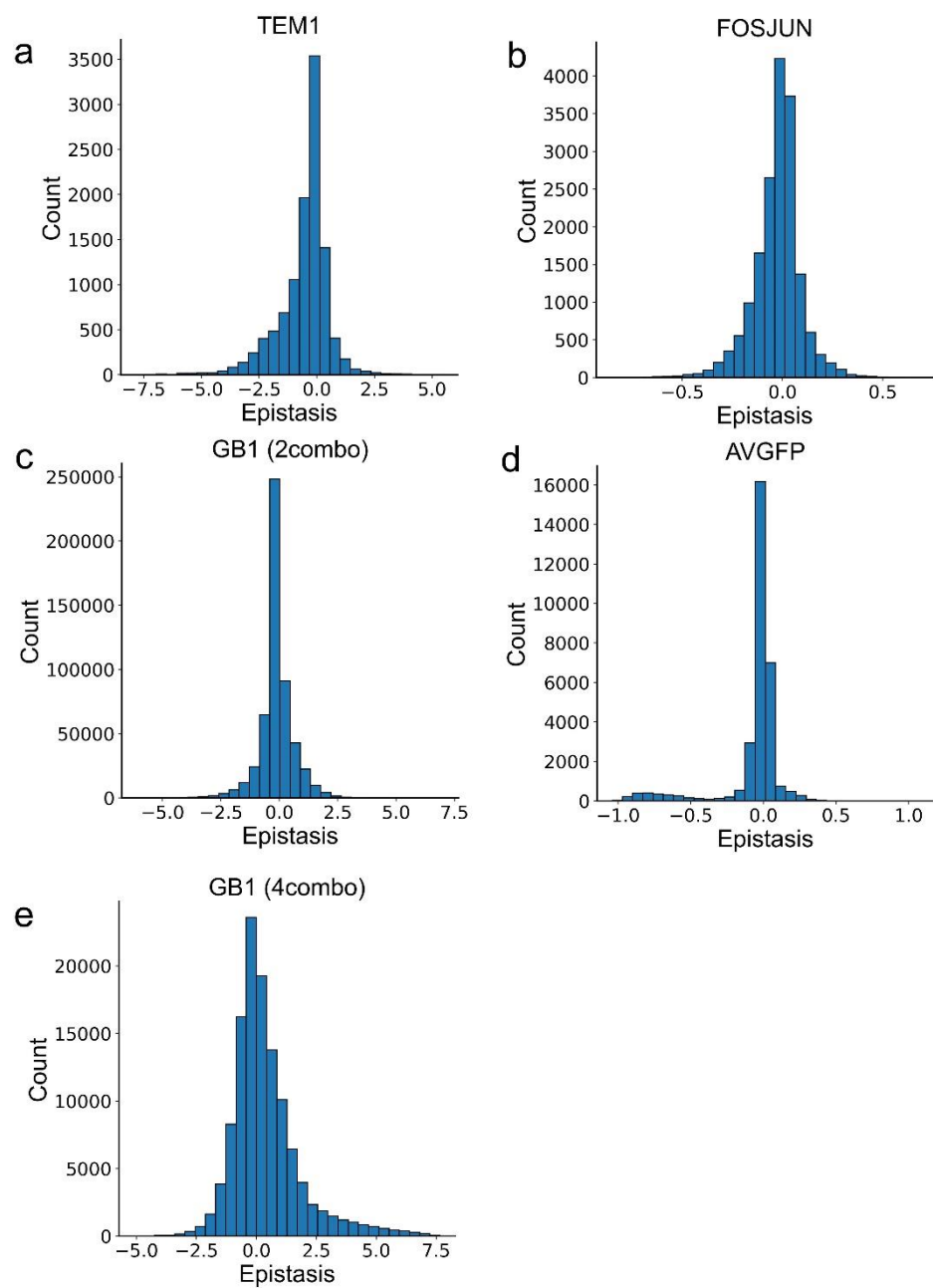


Figure S4. The distribution of epistasis effects of higher-order mutants. Histogram shows the distribution of epistasis value of dataset TEM1 (a), FOSJUN (b), GB1 (2combo) (c), AVGFP (d) and GB1 (4combo) (e).

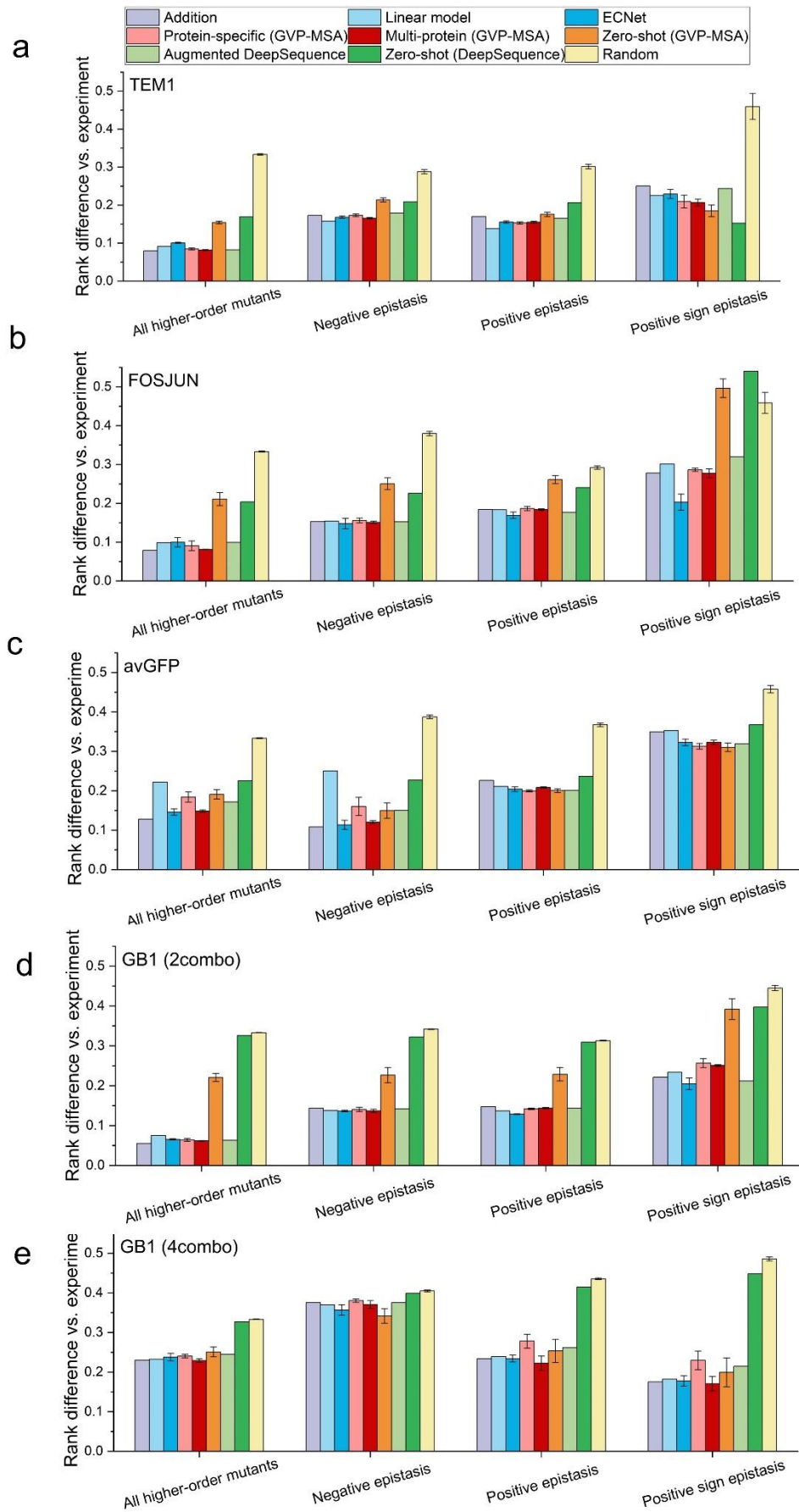


Figure S5. The ranking performance of different DMS datasets when extrapolating to high-order mutation effects from single mutation effects. Comparisons of the mean absolute error (MAE) of rank difference between predictive results and experimental fitness in different groups of mutants. The results in the “Random” method were obtained by averaging 500 random rankings. Lower rank differences indicate better ranking performance.